D05/4/241. Let $f: R^{+} \longrightarrow R$ be a differentiable function satisfying

$$f(xy) = \frac{f(x)}{y} + \frac{f(y)}{x} \quad \forall\ x, y \in R^{+} \qquad \text{also} \qquad f(1) = 0\ ;\ f'(1) = 1$$

find $\lim\limits_{x \to e}\left[\frac{1}{f(x)}\right]$ (where [.] denotes greatest integer function). _________.

माना $f: R^{+} \longrightarrow R$ एक अवकलनीय फलन है जो $f(xy) = \frac{f(x)}{y} + \frac{f(y)}{x}\ \forall\ x, y \in R^{+}$ को संतुष्ट करता है तथा

$f(1) = 0\ ;\ f'(1) = 1$

$\lim\limits_{x \to e}\left[\frac{1}{f(x)}\right]$ ज्ञात कीजिए। (जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है) _________.

Ans. 2

Sol.

$$f'(x) = \underset{h \to 0}{\text{Lt}} \frac{f(x+h) - f(x)}{h} = \underset{h \to 0}{\text{Lt}} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h} = \underset{h \to 0}{\text{Lt}} \frac{\frac{f(x)}{1+\frac{h}{x}} + \frac{f\left(1+\frac{h}{x}\right)}{x} - f(x)}{h}$$

$$= \underset{h \to 0}{\text{Lt}} \frac{f(x)\left(-\frac{h}{x}\right)}{h\left(1+\frac{h}{x}\right)} + \frac{f\left(1+\frac{h}{x}\right)}{hx} = \frac{-f(x)}{x} + \underset{n \to 0}{\text{Lt}} \frac{f\left(1+\frac{h}{x}\right) - f(1)}{x^2\left(\frac{h}{x}\right)} \text{ (as } f(1) = 0)$$

$$f'(x) = \frac{-f(x)}{x} + \frac{f'(1)}{x^2}$$

$$xf'(x) + f(x) = \frac{1}{x}$$

$$\frac{d}{dx}\left(xf(x)\right) = \frac{1}{x}$$

$$xf(x) = \int \frac{1}{x}\,dx$$

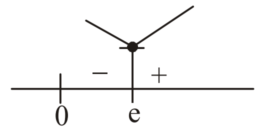
$xf(x) = \ln x + k$

put $x = 1$, weget $k = 0$

$$\therefore f(x) = \frac{\ln x}{x}$$

$$H(x)=\frac{1}{f(x)}=\frac{x}{\ln x}$$

$$H'(x)=H'(x)=\frac{\ln x\cdot 1-1}{(\ln x)^2}$$ $H(x) \ge e$



$H(e) = e$

$$\therefore \underset{x\to e}{\text{Lt}}\left[\frac{1}{f(x)}\right]=2$$

D05/4/242. If $f: R \to R^+$ is differentiable function such that $|f(y)\cdot f(x-y)+(x-y)^2 \ge |f(x)\, f(x-y)|\ \forall\ x \in R$, $f(0) = 5$, then the value of f(5) is ________.

यदि $f: R \to R^+$ एक अवकलनीय फलन इस प्रकार है कि $|f(y)\cdot f(x-y)+(x-y)^2 \ge |f(x)\, f(x-y)|\ \forall\ x \in R$, $f(0) = 5$, तो f(5) का मान होगा ________.

Ans. 5

Sol. $|f(x)\, f(x-y)| - |f(y)\, f(x-y)| \le (x-y)^2$

$\Rightarrow |f(x-y)|\,(|f(x)| - |f(y)| \le (x-y)^2$

$\Rightarrow f(x-y)\,(f(x)-f(y)) \le (x-y)^2$ $(\because\ f(x) \in R^+)$

Case-I : $x > y$

$$f(x-y)\left\{\frac{f(x)-f(y)}{x-y}\right\}\le\{x-y\}$$

$$\Rightarrow \underset{y\to x}{\text{Lim}}\, f(x-y)\left\{\frac{f(y)-f(x)}{y-x}\right\}\le \underset{y\to x}{\text{Lim}}\, f(x-y)$$

$\Rightarrow f(0).\ Lf'(x) \le 0$ ($Lf'(x)$ denotes LHD of f(x))

$\Rightarrow Lf'(x) \le 0$

Case-II : $x < y$

$$f(x-y)\left\{\frac{f(x)-f(y)}{x-y}\right\}\ge\{x-y\}$$

$$\Rightarrow \underset{y\to x}{\text{Lim}}\, f(x-y)\left\{\frac{f(y)-f(x)}{y-x}\right\}\ge \underset{y\to x}{\text{Lim}+}\, f(x-y)$$

$\Rightarrow f(0).Rf'(x) \ge 0$

$\Rightarrow Rf'(x) \ge 0$ ($Rf'(x)$ denotes RHD of f(x))

$\because$ f(x) is differentiable everywhere

$\therefore f'(x) = 0$

$\Rightarrow f(x) = \text{constant}$

$\Rightarrow f(5) = 5 \qquad (\because\ f(0) = 5)$

D05/4/243. If number of points of discontinuity of the function $f(x) = [2 + 15\cos x],\ x \in \left(0, \frac{\pi}{2}\right]$ is same as number of points of non-differentiability of the function $g(x) = [n + m\sin x], x(0, \pi), n, m \in I$ then find the value of m.

Note : [y] denotes largest integer less than or equal to y.

यदि फलन $f(x) = [2 + 15\cos x],\ x \in \left(0, \frac{\pi}{2}\right]$ के असतततता बिन्दुओं की संख्या फलन $g(x) = [n + m\sin x], x(0, \pi), n,$ $m \in I$ के अन–अवकलनीयता बिन्दुओं की संख्या के समान है तो m का मान ज्ञात कीजिए।

नोट : [y], महत्तम पूर्णांक $\le y$ को दर्शाता है।

Ans. 8

Sol. Clearly number of points of $g(x) = [n + m\sin x]$ in $(0, \pi)$ will be $(2m - 1)$

$\therefore \quad 2m - 1 = 15 \quad \Rightarrow m = 8$

D05/4/244. Let f(x) be a polynomial function satisfying $f(x) + f(y) = f(x) \times f(y) - f(xy) + f(1)\ \forall\ x, y \in R$ and $f'(1) = 5$. If number of divisors of $f'(1) \times f''(1) \times f'''(1)$ is N then find the value of $\left(\frac{N}{8}\right)$ _________.

माना f(x), $f(x) + f(y) = f(x) \times f(y) - f(xy) + f(1)\ \forall\ x, y \in R$ को संतुष्ट करने वाला एक बहुपद है तथा $f'(1) = 5$ है। $f'(1) \times f''(1) \times f'''(1)$ के भाजकों की संख्या N है तो $\left(\frac{N}{8}\right)$ का मान ज्ञात कीजिए _________.

Ans. 5

Sol. Putting $y = \frac{1}{x}$, we get

$$f(x)x = +f\left(\frac{1}{x}\right) = f(x) \times f\left(\frac{1}{x}\right)$$

$\Rightarrow \quad f(x) = 1 \pm x^n$

$\Rightarrow \quad f'(x) = \pm n \times x^{n-1}$

$\Rightarrow \quad f'(1) = \pm n = 5$

$\Rightarrow \quad n = 5$ (–ve sign rejected)

$\therefore \quad f(x) = 1 + x^5$

$f'(1) \times f''(1) \times f'''(1) = 5 \times 5 \times 4 \times 5 \times 4 \times 3 = 5^3 \times 2^4 \times 3^1$

$\therefore$ $N = 4 \times 5 \times 2 = 40$

D05/4/245. If $f(x) = \begin{cases} (\operatorname{sgn}(x-2))[\ln x], & 1 \le x \le 3 \\ \{x^2\}, & 3 < x \le \dfrac{7}{2} \end{cases}$

where $[\cdot]$ denotes greatest integer function and $\{\cdot\}$ denotes fraction part function. Then the number of points in interval $\left[1, \dfrac{7}{2}\right]$ where f(x) is discontinuous is ________.

यदि $f(x) = \begin{cases} (\operatorname{sgn}(x-2))[\ln x], & 1 \le x \le 3 \\ \{x^2\}, & 3 < x \le \dfrac{7}{2} \end{cases}$

जहाँ $[\cdot]$ महत्तम पूर्णांक फलन को दर्शाता है तथा $\{\cdot\}$ भिन्नात्मक भाग फलन को दर्शाता है। तब अंतराल $\left[1, \dfrac{7}{2}\right]$ में बिन्दुओं की संख्या जहाँ f(x) असतत् है, होगी _________.

Ans. 5

Sol. for $1 \le x < e$ $\quad [\ln x] = 0$

for $e \le x \le 3$ $\quad [\ln x] = 1$

$\therefore$ f(x) is discontinuous at $x = e$ $\quad (\because$ L.H.L = 0 ; R.H.L = 1)

f(x) is discontinuous at $x = 3$ $\quad (\because$ L.H.L = 1 ; R.H.L = 0)

Also $\{x^2\}$ is discontinuous at $x^2 = 10, 11, 12$ i.e., $x = \sqrt{10}, \sqrt{11}, \sqrt{12}$

$\therefore$ Total number of points of discontinuity = 5

D05/4/246. Let $f_n(x) = \cos^{2n}x$ and $g(x) = \lim\limits_{n \to \infty} \sum\limits_{k=0}^{n} f_k\left(\dfrac{x}{4}\right)$. If g(x) is continuous in (0, c), then the largest value of $\left[\dfrac{c}{2}\right]$.

(where [.] denote greatest integer function) _________.

माना $f_n(x) = \cos^{2n}x$ तथा $g(x) = \lim\limits_{n \to \infty} \sum\limits_{k=0}^{n} f_k\left(\dfrac{x}{4}\right)$ है। यदि g(x), (0, c) में सतत् है, तो $\left[\dfrac{c}{2}\right]$ का महत्तम मान होगा

(जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है) __________.

Ans. 6

Sol. $g(x)=\dfrac{1}{1-\cos^2\dfrac{x}{4}}$

It is continuous in $(0, 4\pi)$

$\therefore \quad c = 4\pi \Rightarrow \left[\dfrac{c}{2}\right]=6$

D02/4/247. The number of solutions of the equation $\cos^{-1}\left(\dfrac{1-x^2-2x}{(x+1)^2}\right)=\pi(1-\{x\})$, for $x \in [0, 76]$ is equal to.

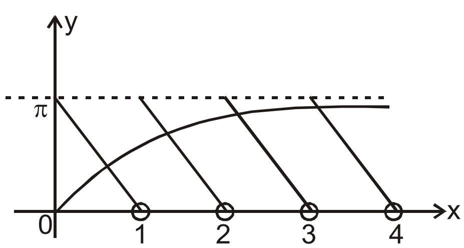
(where $\{\cdot\}$ denote fraction part function) _________.

समीकरण $\cos^{-1}\left(\dfrac{1-x^2-2x}{(x+1)^2}\right)=\pi(1-\{x\})$ के हलों की संख्या क्या होगी, यदि $x \in [0, 76]$

(जहाँ $\{\cdot\}$ भिन्नात्मक भाग फलन है) _________.

Ans. 76

Sol. $\cos^{-1}\left(\dfrac{2}{(1+x)^2}-1\right)=\pi\,(1-\{x\})$



there are total 76 solution.

D02/4/248. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. Given that $f(x) = 0$ has two prime numbers as roots and $b + c = 35$. If the least value of $f(x)\ \forall\ x \in R$ is λ, then $\left[\left|\dfrac{\lambda}{3}\right|\right]$ is equal to

(where [.] denote greatest integer function) _________.

दिया गया है कि $f(x) = x^2 - bx + c$, जहाँ b एक विषम धनात्मक पूर्णांक है। यह दिया गया है कि $f(x) = 0$ के दोनों मूल अभाज्य संख्याएँ हैं तथा $b + c = 35$ है। यदि $f(x)$ का न्यूनतम मान λ है जहाँ $x \in R$ है, तो $\left[\left|\dfrac{\lambda}{3}\right|\right]$ का मान क्या होगा ?

(जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है) _________.

Ans. 6

Sol. $f(x) = x^2 - bx + c = 0 \begin{cases} \nearrow p_1 \\ \searrow p_2 \end{cases}$

$p_1 + p_2 = b$ (odd no.)

$\Rightarrow \quad p_1 = 2$

$p_1p_2 = c$

$b + c = (p_2 + 2) + 2p_2 = 35$

$\Rightarrow \quad p_2 = 11$

$\Rightarrow \quad f(x) = x^2 - 13x + 22$

$\lambda = f(x)_{min} = -\frac{81}{4}$

D02/4/249. Let f(x) be continuous function such that f(0) = 1 and $f(x) - f\left(\frac{x}{7}\right) = \frac{x}{7} \ \forall x \in R$, then f(42) = _________.

यदि f(x) एक सतत् फलन इस प्रकार है कि f(0) = 1 तथा $f(x) - f\left(\frac{x}{7}\right) = \frac{x}{7} \ \forall x \in R$, तो f(42) का मान क्या होगा ?

Ans. 8

Sol. $f'(x) = \lim_{x \to 0} \frac{f(x) - f\left(\frac{x}{7}\right)}{x - \frac{x}{7}} = \frac{1}{6}$

$\Rightarrow \quad f(x) = \frac{x}{6} + 1 \quad \Rightarrow \quad f(42) = 8$

D02/4/250. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\left[f(t) : 0 \le x \le 1\right] & ; \ 0 \le x \le 1 \\ 3 - x & ; \ 1 < x \le 2 \end{cases}$ and if

$\lambda = g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$, then 2λ = _________.

यदि $f(x) = 4x^3 - x^2 - 2x + 1$ तथा $g(x) = \begin{cases} \min\left[f(t) : 0 \le x \le 1\right] & ; \; 0 \le x \le 1 \\ 3 - x & ; \; 1 < x \le 2 \end{cases}$ और यदि

$\lambda = g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$, तो 2λ का मान क्या होगा ?

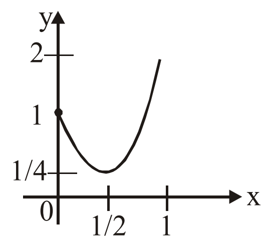
Ans. 5

Sol.

$$g(x) = f(x) \qquad 0 \le x < \frac{1}{2}$$

$$= \frac{1}{4} \qquad \frac{1}{2} \le x \le 1$$

$$= 3 - x \qquad 1 < x \le 2$$



D02/4/251. Let $f(x) = \frac{ax+b}{cx+d}$, where a, b, c, d are non zero. If f(7) = 7, f(11) = 11 and f(f(x)) = x for all x except $-\frac{d}{c}$.

The unique number which is not in the range of f is _________.

यदि $f(x) = \frac{ax+b}{cx+d}$, जहाँ a, b, c, d अशून्य संख्याएँ हैं। यदि f(7) = 7, f(11) = 11 और f(f(x)) = x सभी x के लिए सत्य है, $-\frac{d}{c}$ को छोड़कर तो वह अद्वितीय संख्या बताइए जो f के परिसर में नहीं है _________.

Ans. 9

Sol. f(x) = x has two real roots

$$cx^2 + (d-a)x - b = 0 \begin{matrix} \nearrow 7 \\ \searrow 11 \end{matrix}$$

$$\frac{a-d}{c} = 18 \text{ and } \frac{-b}{c} = 77$$

if $f(f(x)) = x \; \forall \; x \in R \Rightarrow (ac + cd)x^2 + (d^2 - a^2)x - (a + d)b = 0 \Rightarrow a + d = 0 \Rightarrow a = -d$

f(x) will not attain the value $\frac{a}{c} = 9$.

D02/4/252. Let $A = \{x \mid x^2 - 4x + 3 < 0, x \in R\}$

$B = \{x \mid 2^{1-x} + p \le 0; x^2 - 2(p + 7)x + 5 \le 0\}$

If $A \subseteq B$ then the range of real number $p \in [a, b]$ where a, b are integers.

Find the value of (b – a). _________.

माना $A = \{x \mid x^2 - 4x + 3 < 0, x \in R\}$

$B = \{x \mid 2^{1-x} + p \le 0; x^2 - 2(p+7)x + 5 \le 0\}$

यदि $A \subseteq B$ है तो वास्तविक संख्या p का परिसर [a, b] है जहाँ a, b पूर्णांक संख्या हैं।

तो (b – a) का मान बताओ।

Ans. 3

Sol. $A = (1, 3)$

$p \le -2^{1-1}, p \le -2^{1-3}$

$1 - 2(p+7) + 5 \le 0$ and $9 - 6(p+7) + 5 \le 0 \Rightarrow p \in [-4, -1]$

D02/4/253. If $[x] - \{x\} = 3x$, where [], {} represents greatest integer function and fractional part function respectively, then value of $|6\lambda|$, where λ is sum of values of x satisfying it, is _________.

यदि $[x] - \{x\} = 3x$ है जहां [], {} क्रमशः महत्तम पूर्णांक फलन तथा भिन्नात्मक भाग फलन को दर्शाता है। इस समीकरण को सन्तुष्ट करने वाले x के मानों का योग λ है तो $|6\lambda|$ का मान होगा _________.

Ans. 3

Sol. Case I when $x \in I$ equation becomes

$x - 0 = 3x$

$\therefore x = 0$

Case II when $x \notin I \Rightarrow x = m + f$ equation becomes

$m - f = 3m + 3f$

$4f = -2m \Rightarrow f = \frac{-2m}{4} = -\frac{m}{2}$

$\therefore 0 < f < 1 \Rightarrow 0 < -\frac{m}{2} < 1 \Rightarrow -2 < m < 0$

$\therefore m = -1$ and $f = \frac{1}{2} \Rightarrow x = -\frac{1}{2}$

Now sum of values of x is equal to $-\frac{1}{2}$

$\therefore |6\lambda| = \left|6 \times \left(-\frac{1}{2}\right)\right| = 3$

D02/4/254. The number of solutions of the equation $\log_{[x]}(5 - [x]) = |x - 4|$ is (where $[\cdot]$ denotes greatest integer function). _________.

यदि समीकरण $\log_{[x]}(5-[x]) = |x-4|$ के हलों की संख्या क्या होगी (जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है) _________.

Ans. 3

Sol. For the domain of $\log_{[x]}(5-[x])$

$1 < [x] < 5 \Rightarrow x \in [2, 5)$

If $x \in [2, 3)$ then given equation becomes

$|x-4| = \log_2 3 \Rightarrow \boxed{x = 4 - \log_2 3}$

If $x \in [3, 4) \Rightarrow |x-4| = \log_3 2 \Rightarrow \boxed{x = 4 - \log_3 2}$

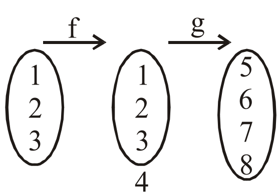
If $x \in [4, 5) \Rightarrow |x-4| = 0 \Rightarrow \boxed{x = 4}$

D02/4/255. Let $f : \{1, 2, 3\} \to \{1, 2, 3\}$ and $g\{1, 2, 3, 4\} \to \{5, 6, 7, 8\}$ be two functions. Let $h(x) = g(f(x))$ such that $h(x)$ is a one-one function. If total number of function $y = g(x)$ is'x' then $\left[\frac{x}{32}\right]$ is equal to ([ ] is GIF) _________.

माना $f : \{1, 2, 3\} \to \{1, 2, 3\}$ तथा $g\{1, 2, 3, 4\} \to \{5, 6, 7, 8\}$ दो फलन हैं। यदि $h(x) = g(f(x))$ इस प्रकार है कि $h(x)$ एकैकी फलन है। यदि $y = g(x)$ के फलनों के फलनों की संख्या 'x' है तो $\left[\frac{x}{32}\right]$ का मान बताओ ([ ] महत्तम पूर्णांक फलन है)

Ans. 3

Sol.



$h(x) = gf(x)$ is one-one

$x = (4 \times 3 \times 2 \times 4)$

$x = 96$

$\left[\frac{x}{32}\right] = 3$

D02/4/256. Let $f : [1, \infty) \longrightarrow R$ be the function defined by $f(x) = \frac{\sqrt{[x]} + \sqrt{\{x\}}}{\sqrt{x}}$

where [.] and {.} denote greatest integer function and fractional part function respectively find minimum value of f(x). _________.

मान लें कि फलन $f : [1, \infty) \longrightarrow R$ निम्नलिखित रूप में परिभाषित है :

$$f(x) = \frac{\sqrt{[x]} + \sqrt{\{x\}}}{\sqrt{x}}$$

जहाँ [.] महत्तम पूर्णांक फलन है और {.} भिन्नात्मक भाग फलन है तो f(x) का न्यूनतम मान ज्ञात करें। _________.

Ans. 1

Sol. $y^2 = \frac{[x] + \{x\} + 2\sqrt{[x].\{x\}}}{x} = 1 + \frac{2\sqrt{[x].\{x\}}}{x} \geq 1$

D02/4/257. If range of $f(x) = \frac{(\ell nx)(\ell nx^2) + \ell nx^3 + 3}{\ell n^2 x + \ell nx^2 + 2}$ can be expressed as $\left[\frac{a}{b}, \frac{c}{d}\right]$ where a, b, c and d are prime numbers (not necessarily distinct) then find the value of $\frac{(a+b+c+d)}{2}$ _________.

यदि $f(x) = \frac{(\ell nx)(\ell nx^2) + \ell nx^3 + 3}{\ell n^2 x + \ell nx^2 + 2}$ का परिसर $\left[\frac{a}{b}, \frac{c}{d}\right]$ है जहाँ a, b, c और d अभाज्य संख्याएँ हैं (जो अनिवार्य रूप से भिन्न नहीं है) तो $\frac{(a+b+c+d)}{2}$ का मान ज्ञात करें।

Ans. 6

Sol. Let ln x = t

$$y = \frac{2t^2 + 3t + 3}{t^2 + 2t + 2} \Rightarrow (y-2)t^2 + (2y-3)t + (2y-3) > 0$$

$$D \geq 0 \Rightarrow (2y-3)(2y-5) \leq 0 \Rightarrow \frac{3}{2} \leq y \leq \frac{5}{2}$$

D02/4/258. Let $A = \{1, 2, 3, 4, 5, 6\}$ be a set. Let $f : A \to A$ be a function such that $f(1) = 2$ and $f(3) = 4$ then number of onto functions $y = f(x)$ such that $f(x) \neq x$ is N. Then $\left[\frac{N}{3}\right] =$ ([.] denotes greatest integer function) _________.

मान लें कि $A = \{1, 2, 3, 4, 5, 6\}$ एक समुच्चय है। एक फलन $f : A \to A$ दिया गया है जिसके लिए $f(1) = 2$ और $f(3) = 4$ है। $y = f(x)$ (जहाँ $f(x) \neq x$) के आच्छादक फलनों की संख्या N है तो $\left[\frac{N}{3}\right]$ का मान ज्ञात करें। (जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है)

Ans. 4

Sol. Total number of functions = 4 ! = 24

Case-I : $f(5) = 5$ and $f(6) = 6$

Total number of ways = $2 \times 1 = 2$

Case-II : $f(5) = 5$ but $f(6) \neq 6$ or $f(6) = 6$ but $f(5) \neq 5$

total number of ways = $2\,[2 \times 2] = 8$

Required number of functions = $24 - 8 - 2 = 14 = N$

D02/4/259. If $x = \log_4\left(\frac{2f(x)}{1-f(x)}\right)$, then find $f(2017) + f(-2016)$ _________.

यदि $x = \log_4\left(\frac{2f(x)}{1-f(x)}\right)$ है तो $f(2017) + f(-2016)$ का मान होगा।

Ans. 1

Sol. $f(x) = \frac{4^x}{2+4^x}$

$f(1-x) + f(x) = 1$

D02/4/260. Let $y = f(x)$ be a function such that $f(x) + f\left(\frac{2x-3}{x-2}\right) = x$, then number of integers in the domain of f(x) is/are _________.

माना y = f(x) एक फलन इस प्रकार है कि $f(x)+f\left(\frac{2x-3}{x-2}\right)=x$, तो f(x) के प्रांत में पूर्णांकों की संख्या होगी।

Ans. 2

Sol. $f(x)+f\left(\frac{2x-3}{x-2}\right)=x$ ...(1)

Put $x \to \frac{2x-3}{x-2}$

$f\left(\frac{2x-3}{x-2}\right)+f(x)=\frac{2x-3}{x-2}$ ...(2)

From (1) and (2)

$x=\frac{2x-3}{x-2}$ $\Rightarrow x^2-4x+3=0 \Rightarrow x=3$ or 1

D02/4/261. If f is a function defined on the set of non-negative integers and taking values in the same set, given that

(i) $x-f(x)=19\left[\frac{x}{19}\right]-90\left[\frac{f(x)}{90}\right]$

(ii) 1900 < f(1990) < 2000

Find unit digits of sum of possible values of f(1990) (where [.] denotes the greatest integer function) _________.

यदि फलन f अऋणात्मक पूर्णांक मानों के समुच्चय में परिभाषित है तथा यह इसी समुच्चयों में मान इस प्रकार रखता है कि

(i) $x-f(x)=19\left[\frac{x}{19}\right]-90\left[\frac{f(x)}{90}\right]$

(ii) 1900 < f(1990) < 2000

f(1990) के सम्भावित मानों के योग का इकाई अंक ज्ञात करो (जहां [.] महत्तम पूर्णांक फलन को दर्शाता है) _________.

Ans. 8

Sol. Since 1900 < f(1990) < 2000

Therefore $\frac{1900}{90}<\frac{f(1900)}{90}<\frac{2000}{90}$

$$\Rightarrow \left[\frac{1900}{90}\right] < \left[\frac{f(1990)}{90}\right] < \left[\frac{2000}{90}\right]$$

$$\Rightarrow 21 < \left[\frac{f(1990)}{90}\right] < 22 \begin{Bmatrix} |21:11| = 21 \\ |22:22| = 22 \end{Bmatrix}$$

Case I : Suppose $= \left[\frac{f(1990)}{90}\right] = 21$ then substituting $x = 1990$ in equation (1)

$$1990 - f(1990) = 19 - 19\left[\frac{1990}{90}\right] - 90\left[\frac{f(1990)}{90}\right]$$

$\Rightarrow f(1990) = 1904$

Case II : Suppose $\left[\frac{f(1990)}{90}\right] = 22$ then substituting $x = 1990$ in equation (1)

$f(1990) = 1904$

D02/4/262. If $g(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{\frac{1}{7}}$, then the value of $g(g(9))$ is equal to ________.

यदि $g(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{\frac{1}{7}}$ है तो $g(g(9))$ का मान होगा ________.

Ans. 9

Sol. We have $4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7 = 4\cos^4 x - 2(2\cos^2 x - 1) - \frac{1}{2}(2\cos^2 2x - 1) - x^7$

$$= 4\cos^4 x - 4\cos^2 x + 2 - (2\cos^2 x - 1)^2 + \frac{1}{2} - x^7 = \left(\frac{3}{2} - x^7\right)$$

We get $g(x) = \left(\frac{3}{2} - x^7\right)^{\frac{1}{7}}$

$$\Rightarrow g(g(x)) = \left(\frac{3}{2} - (g(x))^7\right)^{\frac{1}{7}} = \left(\frac{3}{2} - \left(\frac{3}{2} - x^7\right)\right)^{\frac{1}{7}} = x$$

Hence $g(g(100)) = 100$

D02/4/263. Let $f(x) = [\sec\{x\}]$ where $[x]$ and $\{x\}$ denote greatest integer and fractional part of x respectively and $g(x) = 2x^2 - 3x(K + 1) + K(3K + 1)$. Find number of integral values of K if $g(f(x)) < 0\ \forall\ x \in R$. _________.

मान लें कि $f(x) = [\sec\{x\}]$ जहाँ $[x]$ महत्तम पूर्णांक फलन को और $\{x\}$ भिन्नात्मक भाग फलन को दर्शाता है और $g(x) = 2x^2 - 3x(K + 1) + K(3K + 1)$ है। तो उन पूर्णांक K के मानों की संख्या होगी यदि $g(f(x)) < 0\ \forall\ x \in R$. _________.

Ans. 1

Sol. $0 \le \{x\} < 1$

$\Rightarrow \quad 1 \le \sec\{x\} < 2$

$\Rightarrow \quad [\sec\{x\}] = 1$

$g(f(x)) < 0 \quad \Rightarrow \quad g(1) < 0$

$\Rightarrow \quad 3k^2 - 2k - 1 < 0 \qquad \Rightarrow \qquad -\frac{1}{3} < k < 1$

D02/4/264. Let a function f is defined as $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$. If f satisfy $f(f(x)) = f(x)$, $\forall\ x \in \{1, 2, 3, 4, 5\}$ If number of such functions are k^2 then $k/2$ is _________.

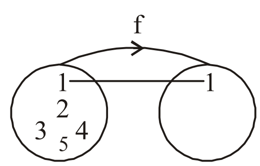
माना एक फलन f इस प्रकार परिभाषित है कि $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$. यदि f, $f(f(x)) = f(x)$ को सन्तुष्ट करता है $\forall\ x \in \{1, 2, 3, 4, 5\}$. यदि ऐसे कुल फलनों की संख्या k^2 है तब $k/2$ होगा :

Ans. 7

Sol. $f(f(x)) = f(x)$; $f(x) = y \Rightarrow \quad f(y) = y$

Case-1: range contains exactly one element

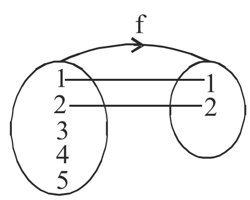
it can be done in $^5C_1$ ways say 1



remaining 4 elements i.e. 2, 3, 4, 5 can be mapped only in one ways $\Rightarrow$ total $= {}^5C_1 \cdot 1 = 5$

Case-2: range contains two elements

this can be done in $^5C_2$ ways say 1, 2

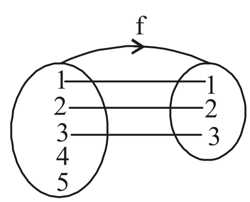


$\therefore \quad f(1) = 1;\ f(2) = 2$

remaining 3 elements i.e. 3, 4 and 5 each can be mapped in 2 ways

Total $= {}^5C_2 \cdot 2^3 = 80$

Case-3: range contains 3 elements which can be done in $^5C_3$ ways say 1, 2, 3

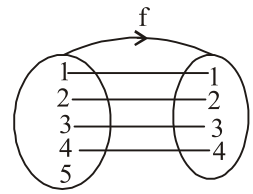


$\therefore \quad f(1) = 1;\ f(2) = 2$ and $f(3) = 3$

now remaining 4 and 5 can be mapped only in 3 ways

Total $= {}^5C_3 \cdot 3^2 = 90$ ways

Case-4: range contains all 4 elements which can be done in ${}^5C_4$ ways say 1, 2, 3, 4



f (1) = 1; f (2) = 2 ; f (3) = 3; f (4) = 4

Now remaining 5 can be mapped in 4 ways

Total = ${}^5C_4 \cdot 4 = 20$

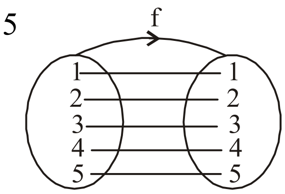
Case-5: range contains all 5 elements which

can be done in ${}^5C_5$ ways say 1, 2, 3, 4, 5

f (1) = 1; f (2) = 2 ; f (3) = 3; f (4) = 4, f (5) = 5

only 1 way

Total = 5 + 80 + 90 + 20 + 1 = 196 = k^2



D02/4/265. Let $f : R \to R$ be a bijective function and the graph of f(x) is symmetric about point (1008, 0) then the sum of digits of the value of $f^{-1}(2016) + f^{-1}(-2016)$ is ________.

माना $f : R \to R$ एकैकी आच्छादी फलन है तथा f(x) का ग्राफ बिन्दु (1008, 0) के सापेक्ष सममित है तो $f^{-1}(2016) + f^{-1}(-2016)$ के मान के अंकों का योग क्या होगा ?

Ans. 9

Sol. $\because$ $F : R \to R$ is bijective and graph of f(x) is symmetric about point (1008, 0)

$\Rightarrow$ $f(1008 + x) + f(1008 - x) = 0\ \forall\ x \in R$

$\therefore$ If $f(1008 + a) = 2016$ then $f(1008 - a) = -2016$

$\Rightarrow$ $f^{-1}(2016) + f^{-1}(-2016) = (1008 + a) + (1008 - a) = 2016$

$\therefore$ sum of digits = 2 + 1 + 6 = 9

D03/4/266. Solve the equation : $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$, x = k.If $f'\left(\frac{1}{2}\right)=6$ & f ′(1) = 4, then

$$\lim_{h\to 0}\frac{f\left(2h+\frac{1}{2}+h^2\right)-f\left(\frac{1}{2}\right)}{f(h-h^2+1)-f(1)}+\frac{1}{k}$$ ________.

समीकरण $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\frac{\pi}{2}$ का हल x = k है। यदि $f'\left(\frac{1}{2}\right)=6$ & f ′(1) = 4 है, तो

$$\lim_{h\to 0}\frac{f\left(2h+\frac{1}{2}+h^2\right)-f\left(\frac{1}{2}\right)}{f(h-h^2+1)-f(1)}+\frac{1}{k}$$ का मान बताओ।

Ans. 6

Sol.

D03/4/267. If $\sum_{n=0}^{\infty} 2\cot^{-1}\left(\frac{n^2+n+4}{2}\right)=k\pi$. Let A be 3×3 matrix given by A = $[a_{ij}]$ and B be a column vector such that B^TAB is a null matrix for every column vector B. If $C = A - A^T$ and $a_{13}=k$, $a_{23}=-5$, $a_{21}=15$, then find the value of det (adj A) + det (adj C). _________.

यदि $\sum_{n=0}^{\infty} 2\cot^{-1}\left(\frac{n^2+n+4}{2}\right)=k\pi$ माना A = $[a_{ij}]$ एक 3 × 3 का आव्यूह है और B एक स्तम्भ सदिश इस प्रकार है कि B^TAB प्रत्येक स्तम्भ सदिश B के लिए शून्य आव्यूह है। यदि $C = A - A^T$ तथा $a_{13}=k$, $a_{23}=-5$, $a_{21}=15$, तब det (adj A) + det (adj C) का मान होगा।

Ans. 0

Sol. $$\sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{2}{n^2+n+4}\right)=\sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{\frac{1}{2}}{\frac{n^2}{4}+\frac{n}{4}+1}\right)=\sum_{n=0}^{\infty} 2\tan^{-1}\left(\frac{\left(\frac{n}{2}+\frac{1}{2}\right)-\frac{n}{2}}{\frac{n}{2}\left(\frac{n}{2}+\frac{1}{2}\right)+1}\right)$$

$$=\sum_{n=0}^{\infty} 2\left(\tan^{-1}\left(\frac{n}{2}+\frac{1}{2}\right)-\tan^{-1}\left(\frac{n}{2}\right)\right)$$

D03/4/268. If $\tan\left(\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r-1}{(r^2+r+1)(r^2-r+1)-2r^3}\right)\right)=961$ then the sum of digit of n is k. If $\sin\theta$ and $\cos\theta$ are roots of $3x^2-2x+\left(p^{\frac{1}{3}}+p^{\frac{2}{3}}+1\right)=0$, and $\sin^k\theta+\cos^k\theta=\frac{p}{q}$ (where p and q are coprime numbers) then value of q – p – 20 is : _________.

यदि $\tan\left(\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r-1}{(r^2+r+1)(r^2-r+1)-2r^3}\right)\right)=961$ है, तो n के अंको का योग k होगा। यदि $\sin\theta$ तथा $\cos\theta$ समीकरण $3x^2-2x+\left(p^{\frac{1}{3}}+p^{\frac{2}{3}}+1\right)=0$ के मूल हो, तथा $\sin^k\theta+\cos^k\theta=\frac{p}{q}$ (जहाँ p और q सहअभाज्य संख्याऐं है) तो q – p – 20 का मान क्या होगा

Ans. 5

Sol. $$\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r-1}{\left(r^2+r+1\right)\left(r^2-r+1\right)-2r^3}\right)$$

$$=\sum_{r=1}^{n}\tan^{-1}\left(\frac{2r-1}{1+r^2\left(r^2-2r+1\right)}\right)$$

$$=\sum_{r=1}^{n}\tan^{-1}\left(\frac{r^2-\left(r-1\right)^2}{1+r^2\left(r-1\right)^2}\right)$$

$$=\sum_{r=1}^{n}\left(\tan^{-1}(r)^2-\tan^{-1}(r-1)^2\right)$$

$= \tan^{-1}(1)^2 - \tan^{-1}(0)^2 + \tan^{-1}(2)^2$ .....

$\tan^{-1}(1)^2 + \tan^{-1}(3)^2 - \tan^{-1}(2)^2$ + ...

$\tan^{-1}(n)^2$

$\tan(\tan^{-1}(n^2)) = 961$

$\Rightarrow n^2 = 961 \Rightarrow n = 31$

D03/4/269. If range of the function $f(x) = \sin^{-1}x + 2\tan^{-1}x + x^2 + 4x + 1$ is $[p, q]$ then the value of $(p + q)$ is k.Let $f\left(n\right)=\frac{kn+\sqrt{kn^2-1}}{\sqrt{2n+1}+\sqrt{2n-1}}$, $n \in N$ then the remainder when $f(1) + f(2) + f(3) + ... + f(60)$ is divided by 9 is. _________.

यदि फलन $f(x) = \sin^{-1}x + 2\tan^{-1}x + x^2 + 4x + 1$ का परिसर $[p, q]$ है तो $(p + q)$ का मान k है। माना $f\left(n\right)=\frac{kn+\sqrt{kn^2-1}}{\sqrt{2n+1}+\sqrt{2n-1}}$, $n \in N$ तो $f(1) + f(2) + f(3) + ... + f(60)$ को 9 से भाग देने पर शेषफल क्या मिलेगा।

Ans. 8

Sol.

$$f(n)=\frac{(2n+1)+(2n-1)+\sqrt{(2n+1)(2n-1)}}{\sqrt{2n+1}+\sqrt{2n-1}}$$

Let $\sqrt{2n+1}=a$ and $\sqrt{2n-1}=b$

$$f(n)=\frac{(a^2+b^2+ab)}{(a+b)}\frac{(a-b)}{a-b}=\frac{a^3-b^3}{a^2-b^2}\Rightarrow f(n)=\frac{(2n+1)^{3/2}-(2n-1)^{3/2}}{2}$$

$$\sum_{n=1}^{60}f(n)=\sum_{n=1}^{60}\frac{(2n+1)^{3/2}-(2n-1)^{3/2}}{2}=\frac{(121)^{3/2}-1}{2}=665$$

D03/4/270. The number of ordered pairs of (x, y), which satisfy $y = |\sin x|$ and $y = \cos^{-1}(\cos x)$is k, where $-2\pi \le x \le 2\pi$. For a triangle ABC is given that, $\cos A + \cos B + \cos C = \frac{3}{2}$. then the value of

$(kA + B - kC)\frac{9}{\pi} + 1$ will be _________.

अन्तराल $-2\pi \le x \le 2\pi$ उन क्रमित युग्मों की संख्या k होगी, जो $y = |\sin x|$ तथा $y = \cos^{-1}(\cos x)$ को सन्तुष्ट कर सके।

एक त्रिभुज ABC में दिया गया है $\cos A + \cos B + \cos C = \frac{3}{2}$, तो $(kA + B - kC)\frac{9}{\pi} + 1$ का मान होगा।

Ans. 4

Sol.

D03/4/271. If $f(x) = \cos^{-1}\cos x$, $f(2009) = a\pi + 2009b$ and a, b are rational numbers, then value of $\frac{a}{16} + 39b$ is k. Centre of the tetrahedron OABC with O being origin and A(a, k, 3), B(2, b, –6), C(–k, 4, c) is (k, 3, –1) then the value of $a^2 - b + 4c^2$ is _________.

यदि $f(x) = \cos^{-1}\cos x$, $f(2009) = a\pi + 2009b$ तथा a, b परिमेय संख्याएँ हैं तो $\frac{a}{16} + 39b$ का मान k है। यदि चतुष्फलक OABC का केन्द्र (k, 3, – 1) है, जहाँ O मूलबिन्दु है और बिन्दु A(a, k, 3), B(2, b, –6), C(–k, 4, c) हैं तो $a^2 - b + 4c^2$ का मान ज्ञात करें। _________.

Ans. 6

Sol. $\frac{0 + a + 2 + (-1)}{4} = 1 \Rightarrow a = 3$

$\frac{0 + 1 + b + 4}{4} = 3 \Rightarrow b = 7$

$\frac{0 + 3 - 6 + c}{4} = -1 \Rightarrow c = -1$

$a^2 - b + 4c^2 = 6$

D03/4/272. Find the number of integer(s) in the range of $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)\ \forall\ x \in R$ is k. ABCD is a tetrahedron with pv's of its angular points as A(–5, 22, 5); B(1, 2, 3); C(k, 3, 2) and D(–1, 2, – 3). If the area of the triangle AEF where the quadrilaterals ABDE and ABCF are parallelograms is $\sqrt{S}$ then find the value of S. _________.

फलन $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)\ \forall\ x \in R$ के परिसर में आने वाले पूर्णांकों की संख्या k है। ABCD एक चतुष्फलक है, जिसके शीर्षों के निर्देशांक निम्नलिखित हैं A(–5, 22, 5); B(1, 2, 3); C(k, 3, 2) और D(–1, 2, – 3) हैं। यदि त्रिभुज

AEF का क्षेत्रफल $\sqrt{S}$ है जहाँ चतुर्भुज ABDE और ABCF समांतर चतुर्भुज हैं, तो S का मान ज्ञात कीजिए।

Ans. 110

Sol. $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

$$f(x) = \begin{cases} 2x & x \in \left[0, \frac{\pi}{2}\right] \\ \pi & x \in \left[\frac{\pi}{2}, \pi\right] \\ 3\pi - 2x & x \in \left[\pi, \frac{3\pi}{2}\right] \\ 0 & x \in \left[\frac{3\pi}{2}, 2\pi\right] \end{cases}$$

$\therefore$ Range of f(x) is $[0, \pi]$

D03/4/273. In each of a set of games it is 2 to 1 in favour of the winner of the previous game. the chance that the player who wins the first game shall win three at least, of the next four is m/n where m, n are coprime. If $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) = \tan^{-1}\frac{m}{x}$ then the value of $250x^4 + 320x^2 - 1630$ is equal to. _________.

किसी खेलों के समूह में प्रत्येक खेल में पिछले खेल के विजेता के पक्ष में 2 : 1 का अनुपात होता है। पहले खेल को जीतने वाले खिलाड़ी के अगले चार में से कम से कम तीन खेल जीतने की संभावना m/n है जहाँ m, n सह अभाज्य संख्याएँ हैं। यदि $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) = \tan^{-1}\frac{m}{x}$ तो $250x^4 + 320x^2 - 1630$ का मान बताओ।

Ans. 10

Sol. We have

$$\frac{x + \frac{2}{x} - \left(x - \frac{2}{x}\right)}{1 + \left(x^2 - \frac{4}{x^2}\right)} = \frac{4}{x} \quad \Rightarrow \quad x^2 - \frac{4}{x^2} = 0$$

$\Rightarrow \quad x^2 = 2$

So $250x^4 + 320x^2 + 137 = 250 \times 4 + 320 \times 2 - 1630 = 10$

D04/4/274. Let $0 \le \beta_r \le 1$ and $\sum_{r=1}^{k} \cos^{-1}\beta_r = \frac{k\pi}{2}$ for any $k \ge 1$ and $A = \sum_{r=1}^{k} (\beta_r)^r$, then

$$\lim_{x\to A} 12.\frac{\left(1+x^2\right)^{\frac{1}{3}}-\left(1-2x\right)^{\frac{1}{4}}}{x+x^2}$$ is equal to ________.

माना $0 \le \beta_r \le 1$ तथा $\sum_{r=1}^{k} \cos^{-1}\beta_r = \frac{k\pi}{2}\ \forall\ k \ge 1$ तथा $A = \sum_{r=1}^{k}(\beta_r)^r$ हैं, तब

$$\lim_{x\to A} 12.\frac{\left(1+x^2\right)^{\frac{1}{3}}-\left(1-2x\right)^{\frac{1}{4}}}{x+x^2}$$ का मान होगा ________.

Ans. 6

Sol. Since maximum value of $\cos^{-1}x$ in $[0, 1] = \frac{\pi}{2}$, $\sum_{r=1}^{k}\cos^{-1}\beta_r = \frac{k\pi}{2}$ is possible if and only if each $\cos^{-1}\beta_r = \frac{\pi}{2}$

$\Leftrightarrow \beta_r = 0$

$$A = \sum_{r=1}^{k}(\beta_r)^r = 0$$

Thus $\lim_{x\to 0}\frac{\left(1+x^2\right)^{\frac{1}{3}}-\left(1-2x\right)^{\frac{1}{4}}}{x+x^2} = \frac{1}{2}$. Thus the reqd. limit is 6.

D04/4/275. Let $\Delta = \begin{vmatrix} \tan x & \tan(x+h) & \tan(x+2h) \\ \tan(x+2h) & \tan x & \tan(x+h) \\ \tan(x+h) & \tan(x+2h) & \tan x \end{vmatrix}$ and $f(x) = \lim_{h\to 0}\frac{\Delta}{h^2}$ then $\left[f\left(\frac{\pi}{6}\right)\right]$ is

(where [.] greatest integer function) ________.

माना $\Delta = \begin{vmatrix} \tan x & \tan(x+h) & \tan(x+2h) \\ \tan(x+2h) & \tan x & \tan(x+h) \\ \tan(x+h) & \tan(x+2h) & \tan x \end{vmatrix}$ तथा $f(x) = \lim_{h\to 0}\frac{\Delta}{h^2}$ हैं तब $\left[f\left(\frac{\pi}{6}\right)\right]$ होगा

(जहाँ [.] महत्तम पूर्णांक फलन है) ________.

Ans. 9

Sol. $f(x) = 9\tan x \sec^4 x$

$$f\left(\frac{\pi}{6}\right) = \frac{16\sqrt{3}}{3}$$

D04/4/276. Let f(x) be a quadratic polynomial such that $f(-\pi) = f(\pi) = 0$ and $f\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4}$.

If $\lim\limits_{x\to\pi} f(x).\cos(\sin x).\text{cosec}(\sin x) = m\pi$, then find the value of m. _________.

माना f(x) एक द्विघातीय बहुपद इस प्रकार है कि $f(-\pi) = f(\pi) = 0$ तथा $f\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4}$ हैं।

यदि $\lim\limits_{x\to\pi} f(x).\cos(\sin x).\text{cosec}(\sin x) = m\pi$ है, तो m का मान ज्ञात कीजिए _________.

Ans. 2

Sol. $f(-\pi) = f(\pi) = 0$

$\Rightarrow \quad f(x) = a(x^2 - \pi^2)$

If $f\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4}$ then $f(x) = \pi^2 - x^2$

$$\lim_{x\to\pi} \frac{(\pi - x)(\pi + x)\cos(\sin x)}{\sin(\sin x)} = 2\pi$$

D04/4/277. Value of $\lim\limits_{x\to0} \frac{\cos^2\{1-\cos^2(1-\cos^2(............(1-\cos^2(x))))\}}{\sin\left[\pi\left(\frac{\sqrt{x+4}-2}{x}\right)\right]}$ is L, then L^2 is :

सीमा का मान $\lim\limits_{x\to0} \frac{\cos^2\{1-\cos^2(1-\cos^2(............(1-\cos^2(x))))\}}{\sin\left[\pi\left(\frac{\sqrt{x+4}-2}{x}\right)\right]}$ होगा _________.

Ans. 2

Sol. Let $A = \frac{\cos^2\{1-\cos^2(1-\cos^2(............(1-\cos^2(x))))\}}{\sin\left[\pi\left(\frac{\sqrt{x+4}-2}{x}\right)\right]}$

$$= \lim_{x\to 0} \frac{\cos^2\{\sin^2(\sin^2(......(1-\cos^2(x))))\}}{\sin\left(\pi\left(\frac{\sqrt{x+4}-2}{x}.\frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}\right)\right)}$$

$$= \frac{\cos^2 0}{1}\lim_{x\to 0}\frac{1}{\sin\left(\pi\frac{(x+4-4)}{x\left(\sqrt{x+4}+2\right)}\right)} = \frac{1}{\sin\frac{\pi}{4}} = \sqrt{2}$$

D04/4/278. If α, β are two distinct real roots of the equation $ax^3 + x - 1 - a = 0$, $(a \neq -1, 0)$, none of which is equal to unity, then the value of $\lim\limits_{x\to\left(\frac{1}{\alpha}\right)} \frac{(1+a)x^3 - x^2 - a}{\left(e^{1-\alpha x}-1\right)(x-1)}$ is $\frac{a\ell(k\alpha-\beta)}{\alpha}$, find the value of $k\ell$ ________.

यदि α, β समीकरण $ax^3 + x - 1 - a = 0$, $(a \neq -1, 0)$ के दो भिन्न तथा वास्तविक मूल हैं, जिनमें से कोई भी इकाई नहीं है, तब $\lim\limits_{x\to\left(\frac{1}{\alpha}\right)} \frac{(1+a)x^3 - x^2 - a}{\left(e^{1-\alpha x}-1\right)(x-1)}$ का मान $\frac{a\ell(k\alpha-\beta)}{\alpha}$ है, $k\ell$ का मान ज्ञात कीजिए ________.

Ans. 1

Sol. $a(x^3 - 1) + (x - 1) = 0$

$(x - 1)(ax^2 + ax + a + 1) = 0$

$\alpha, \beta \neq 1$ so, α, β are roots of

$ax^2 + ax + a + 1 = 0$

$\alpha + \beta = -1,\ \alpha\beta = \frac{a+1}{a}$

$$= \underset{x\to\frac{1}{\alpha}}{\mathrm{Lt}} \frac{(1+a)x^3 - x^2 - a}{\left(e^{1-\alpha x}-1\right)(x-1)} = \underset{x\to\frac{1}{\alpha}}{\mathrm{Lt}} \frac{\left(x^3-x^2\right)+a\left(x^3-1\right)}{\left(2^{1-\infty x}-1\right)(x-1)}$$

$$= \underset{x\to\frac{1}{\alpha}}{\mathrm{Lt}} \frac{\left|x^2+a\left(x^2+x+1\right)\right|}{\left(e^{1-\alpha x}-1\right)}$$

$$= \underset{x\to\frac{1}{\alpha}}{\mathrm{Lt}} \frac{(1+a)x^2+ax+a}{\left(\frac{e^{1-\alpha x}-1}{1-\alpha x}\right)(1-\alpha x)}$$

$$= \underset{x\to\frac{1}{\alpha}}{\text{Lt}}\ a\frac{\left[\left(\frac{1+a}{a}\right)x^2+(1)x+1\right]}{(1-\alpha x)} = \underset{x\to\frac{1}{\alpha}}{\text{Lt}}\ a\frac{\left(\alpha\beta x^2-(\alpha+\beta)x+1\right)}{(1-\alpha x)} = \underset{x\to\frac{1}{\alpha}}{\text{Lt}}\ a\frac{(1-(\alpha)x)(1-(\beta)x)}{(1-\alpha x)}$$

$$=\frac{a(\alpha-\beta)}{\alpha}$$

D04/4/279. If $f(x)=\lim\limits_{h\to0}\dfrac{(\sin(x+h))^{\ln(x+h)}-(\sin x)^{\ln x}}{h}$ find $f\left(\dfrac{\pi}{2}\right)$ _________.

यदि $f(x)=\lim\limits_{h\to0}\dfrac{(\sin(x+h))^{\ln(x+h)}-(\sin x)^{\ln x}}{h}$ है तो $f\left(\dfrac{\pi}{2}\right)$ का मान ज्ञात कीजिए _________.

Ans. 0

Sol. Let $g(x)=(\sin x)^{\ln x}$

$$f(x)=\lim_{h\to0}\frac{g(x+h)-g(x)}{h}=g'(x)$$

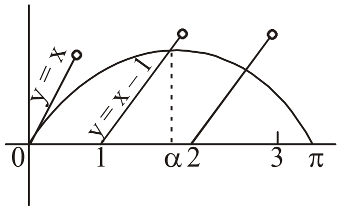
$$f(x)=g'(x)=\frac{d}{dx}\left(e^{\ln x.\ln\sin x}\right)=(\sin x)^{\ln x}\left(\frac{1}{x}.\ln\sin x+\ln x.\frac{1}{\sin x}.\cos x\right)$$

D04/4/280. Find $\lim\limits_{x\to\alpha^+}\left[\dfrac{\min(\sin x,\{x\})}{x-1}\right]$ where α is root of equation $\sin x+1=x$ (here [.] represent greatest integer and {.} represent fractional part function) _________.

$\lim\limits_{x\to\alpha^+}\left[\dfrac{\min(\sin x,\{x\})}{x-1}\right]$ ज्ञात कीजिए जहाँ α समीकरण $\sin x+1=x$ का मूल है (जहाँ [.] महत्तम पूर्णांक फलन तथा {.} भिन्नात्मक भाग फलन को दर्शाता है) _________.

Ans. 0

Sol. $\lim\limits_{x\to\alpha^+}\left[\dfrac{\sin x}{x-1}\right]=0$



D04/4/281. The value of $\lim_{x\to 0}\dfrac{-\sin^3 x + x^3 \,\text{sgn}\left(1-\left[\dfrac{\tan^{-1} x}{x}\right]\right)}{x\tan^2 x\sin(\pi\cos x)}$ is equal to $\dfrac{k}{\pi}$, then $|k| =$ ________.

([.] denotes greatest integer function)

$\lim_{x\to 0}\dfrac{-\sin^3 x + x^3 \,\text{sgn}\left(1-\left[\dfrac{\tan^{-1} x}{x}\right]\right)}{x\tan^2 x\sin(\pi\cos x)}$ का मान $\dfrac{k}{\pi}$ है, तो $|k| =$ ________.

([.] महत्तम पूर्णांक फलन को दर्शाता है)

Ans. 1

Sol. $\underset{x\to 0}{\text{Lt}}\left[\dfrac{\tan^{-1} x}{x}\right] = 0$

$$\underset{x\to 0}{\text{Lt}}\,\text{sgn}\left(1-\left[\dfrac{\tan^{-1} x}{x}\right]\right) = 1$$

$$\underset{x\to 0}{\text{Lt}}\frac{x^3-\sin^3 x}{x.\tan^2 x.\sin(\pi\cos x)} = \underset{x\to 0}{\text{Lt}}\frac{1-\left(\dfrac{\sin x}{x}\right)^3}{\dfrac{\tan^2 x}{x^2}.\sin(\pi-\pi\cos x)}$$

$$= \underset{x\to 0}{\text{Lt}}\frac{1-\left(1-\dfrac{x^2}{3!}+\dfrac{x^4}{5!}......\right)^3}{\dfrac{\tan^2 x}{x^2}.\dfrac{\sin(\pi-\pi\cos x)}{\pi-\pi\cos x}.\dfrac{\pi(1-\cos x)}{x^2}.x^2}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{1-\left(1-\frac{x^2}{2}+\frac{13x^4}{120}......\right)}{\frac{\tan^2 x}{x^2}.\frac{\sin(\pi-\pi\cos x)}{\pi-\pi\cos x}.\frac{\pi(1-\cos x)}{x^2}.x^2} = \frac{1}{\pi}$$

D04/4/282. Let $\lim_{n\to\infty} n\sin\left(2\pi e\lfloor n\right) = k\pi$, where $n \in N$. Find k ________.

माना $\lim_{n\to\infty} n\sin\left(2\pi e\lfloor n\right) = k\pi$, जहाँ $n \in N$ है। k होगा ________.

Ans. 2

Sol. $n\sin 2\pi\left(1+1+\frac{1}{\lfloor 2}+\frac{1}{\lfloor 3}..........\frac{1}{\lfloor N}\right)\lfloor n$

$$n\sin 2\pi\left(1+\frac{1}{n+1}+\frac{1}{(n+1)(n+2)}+..........\frac{1}{(n+1)(n+2)........(N)}\right)$$

(Using $\sin(2n\pi+\theta)=\sin\theta$)

$$= n\left(2\pi\right)\left(\frac{1}{n+1}+\frac{1}{(n+1)(n+2)}+.......\frac{1}{(n+1)(n+2)....N}\right)$$

Using $\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$

$$= 2\pi$$

D04/4/283. Let $L = \lim_{x\to\infty}\left(x\ln x + 2x\ln\sin\left(\frac{1}{\sqrt{x}}\right)\right)$. Find the value of $\left(-\frac{1}{L}\right)$ ________.

माना $L = \lim_{x\to\infty}\left(x\ln x + 2x\ln\sin\left(\frac{1}{\sqrt{x}}\right)\right)$ है। $\left(-\frac{1}{L}\right)$ का मान होगा ________.

Ans. 3

Sol. $\lim_{x\to\infty} x\left(\ln x + 2\ln\left(\sin\frac{1}{\sqrt{x}}\right)\right) = \lim_{x\to\infty} x\left(\ln\left(x.\sin^2\left(\frac{1}{\sqrt{x}}\right)\right)\right)$

$$=\lim_{x\to\infty} x\left(\frac{\ln\left(1+x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1\right)}{x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1}\right)\left(x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1\right)$$

$$=\lim_{x\to\infty} \frac{x}{2}\left(\frac{\ln\left(1+x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1\right)}{x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1}\right)\left(x-2-x\cos\frac{2}{\sqrt{x}}\right)$$

$$=\lim_{x\to\infty} \frac{x}{2}\left(\frac{\ln\left(1+x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1\right)}{x\sin^2\left(\frac{1}{\sqrt{x}}\right)-1}\right)\left(x-2-x\left(1-\frac{2}{x}+\frac{2}{3x^2}....\right)\right)=-\frac{1}{3}$$

D04/4/284. If $L=\lim_{x\to\infty}\left\{x-x^2\log_e\left(1+\frac{1}{x}\right)\right\}$, then 8L ________.

यदि $L=\lim_{x\to\infty}\left\{x-x^2\log_e\left(1+\frac{1}{x}\right)\right\}$ है, तो 8L ________.

Ans. 4

Sol. Let $x=\frac{1}{y};\quad y\to 0$

$$\lim_{y\to 0}\left\{\frac{1}{y}-\frac{1}{y^2}\log_e(1+y)\right\}$$

$$\lim_{y\to 0}\left\{\frac{y-\log_e(1+y)}{y^2}\right\}$$

$$\lim_{y\to 0}\left\{\frac{y-\frac{y^2}{2}+\frac{y^3}{3}...}{y^2}\right\}=\frac{1}{2}=L$$

$$8L=8\times\frac{1}{2}=4$$

D09/4/285. If $f(x)=\begin{cases}\sin(\cos x)+\cos(\sin x) & , \quad x\in\left[-\frac{\pi}{2},0\right]\\ \sin(\lambda\cos x)-x+1 & , \quad x\in\left(0,\frac{\pi}{2}\right]\end{cases}$ has a maximum at x = 0, then number of all possible integral values of $\lambda \in [0, 2\pi]$ is :

यदि $f(x)=\begin{cases}\sin(\cos x)+\cos(\sin x) & , \quad x\in\left[-\frac{\pi}{2},0\right]\\ \sin(\lambda\cos x)-x+1 & , \quad x\in\left(0,\frac{\pi}{2}\right]\end{cases}$ का x = 0 पर उच्चिष्ठ मान हो, तो $\lambda \in [0, 2\pi]$ के सभी संभव पूर्णांक मानों की संख्या होगी :

Ans. 6

Sol. $f(0) = \sin 1 + -1$

$f(0^+) = \sin\lambda + 1$

$f(0) > f(0^+) \Rightarrow \sin\lambda < \sin 1$

$\Rightarrow \lambda = 0, 1, 3, 4, 5, 6$

D09/4/286. Let y = f(x) be a differentiable function such that f(x) = f(4 – x). If y = f(x) has a point of minima at $x=\frac{1}{2}$ and a point of maxima at $x=\frac{3}{2}$ then minimum number of real roots of the equation f(x) . f'(x) = 0 is _________.

माना y = f(x) एक अवकलनीय फलन इस प्रकार है कि f(x) = f(4 – x). If y = f(x), $x=\frac{1}{2}$ पर एक निम्निष्ठ बिन्दु तथा $x=\frac{3}{2}$ पर एक उच्चिष्ठ बिन्दु रखता है तब समीकरण f(x) . f'(x) = 0 के वास्तविक मूलों की न्यूनतम संख्या होगी _________.

Ans. 5

Sol. $f(x) = f(4-x) \Rightarrow f'(x) = -f'(4-x)$

$f'\left(\frac{1}{2}\right)=0 \Rightarrow f'\left(\frac{7}{2}\right)=0$

$f'\left(\frac{3}{2}\right)=0 \Rightarrow f'\left(\frac{5}{2}\right)=0$

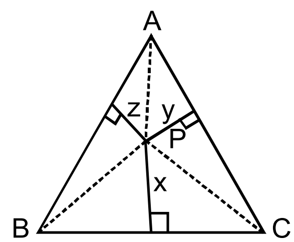
$f'(2) = 0 \Rightarrow f'(x) = 0$ has at least 5 roots.

D09/4/287. Let ΔABC be an equilateral triangle with side length 2010. For any point P inside ΔABC, let d(P) denotes the sum of the distance of P to each of the sides of ΔABC. Let M and m are maximum and minimum values of d(P) respectively. Find (M – m) _________.

माना कि ΔABC एक समबाहु त्रिभुज है जिसकी भुजा की लम्बाई 2010 इकाई है। ΔABC के किसी भी आंतरिक बिन्दु P के लिए माना कि d(P) बिन्दु P की प्रत्येक भुजा से दूरी को प्रदर्शित करता है। माना कि d(P) का अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हो, तब (M – m) ज्ञात करें _________.

Ans. 0

Sol. $\Delta\, ABC = \Delta\, BPC + \Delta\, CPA + \Delta\, APB$



$$\Rightarrow \Delta = \frac{1}{2}ax + \frac{1}{2}ay + \frac{1}{2}az$$

$$\Rightarrow \Delta = \frac{a}{2}(x + y + z)$$

$$\Rightarrow \quad d(P) = x + y + z = \frac{2\Delta}{a} = \text{constant}$$

$$\therefore \quad M = m \Rightarrow M - m = 0$$

D09/4/288. For any 2 squares, the sum of whose area is 1, a rectangle of area A exists into which the squares can be placed without over lapping of interior points (Assume that the squares are to be placed with their sides parallel to the sides of the rectangle). The least value of A can be expressed as $\frac{1+\sqrt{\alpha}}{\beta}$ where α and β are natural numbers, then $(\alpha + \beta)$.

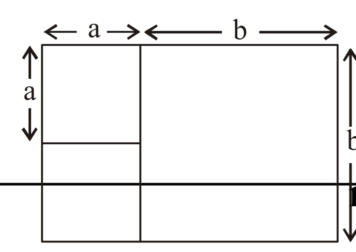
कोई 2 वर्ग जिनके क्षेत्रफलों का योग 1 है, के लिए A क्षेत्रफल का आयत इस प्रकार विद्यमान है कि दोनों वर्ग आयत में रखे जा सके, जबकि दोनों वर्गों में कोई अन्तः बिन्दु उभयनिष्ठ ना हो (मानिये कि वर्गों को इस प्रकार रखा गया है कि उनकी भुजाऐं आयत की भुजाओं के समान्तर हो). यदि A का न्यूनतम मान $\frac{1+\sqrt{\alpha}}{\beta}$ जहाँ $\alpha, \beta \in N$ के रूप में लिखा जा सकता हो, तो $(\alpha + \beta)$ ज्ञात करों।

Ans. 4

Sol. $(b > a)$

$a^2 + b^2 = 1 \qquad (1)$

$A = (a + b)\, b = a\sqrt{1-a^2} + (1-a^2)$



$$\frac{dA}{da}=0 \Rightarrow a^2=\frac{2-\sqrt{2}}{4} \text{ and } b^2=\frac{2+\sqrt{2}}{4} \Rightarrow A=\frac{1+\sqrt{2}}{2}$$

No rain, then $f(x, y)=0$ hence 3 lines

D09/4/289. The equation $2x^3 - 3x^2 + p = 0$ has 3 distinct real roots then [p] is

(Where [.] denotes the greatest integer function).

समीकरण $2x^3 - 3x^2 + p = 0$ के 3 भिन्न वास्तविक मूल है तो [p] का मान होगा

(जहां [.] महत्तम पूर्णांक फलन को प्रदर्शित करता है)

Ans. 0

Sol. Let $f(x) = 2x^3 - 3x^2 + p$

$\Rightarrow f'(x) = 6x^2 . 6x = 0$ at $x = 0, 1$

$\Rightarrow f(0) > 0 \Rightarrow p > 0$

$f(1) < 0 \Rightarrow 2 - 3 + p < 0$

$\Rightarrow p < 1$

$\Rightarrow p \in (0, 1)$

$\Rightarrow [p] = 0$

D09/4/290. $y=f(x)=\frac{x^2-x+1}{x^2+x+1}$. If the area of the quadrilateral bounded by the tangents and normals at the points of maximum and minimum is Δ, then $\frac{3\Delta}{8}$ is ________.

$y=f(x)=\frac{x^2-x+1}{x^2+x+1}$ फलन के अधिकतम तथा न्यूनतम बिन्दुओं पर खींची गई स्पर्श रेखाओं तथा अभिलम्बों से परिबद्ध चतुर्भुज का क्षेत्रफल यदि Δ हो तब $\frac{3\Delta}{8}$ होगा ________.

Ans. 2

Sol. $y=\frac{x^2-x+1}{x^2+x+1}$

$(y - 1)x^2 + (y + 1)x + y - 1 = 0$

$(y + 1)^2 - 4(y - 1)^2 \geq 0$

$(3y - 1) (-y + 3) \geq 0$

$\frac{1}{3} \le y \le 3$

Minimum $x = 1,\ y = \frac{1}{3}$

Maximum $x = -1, y = 3$

Area of the quadrilateral is $= \frac{16}{3} = \Delta$

So $\quad \frac{3\Delta}{8} = 2$

D09/4/291. A variable chord PQ, of inclination θ with x-axis of $y^2 = 4x$ passes through the point R(b, 0). If $f(b,\theta) = \frac{1}{PR^2} + \frac{1}{QR^2}$, then number of point of non differentiability of g(b) = max. f(b, θ) $(0 \le \theta < \pi)$ where b > 0 is _________.

x-अक्ष से θ झुकाव वाली परवलय $y^2 = 4x$ की एक चर जीवा बिन्दु R(b, 0) से गुजरती है। यदि $f(b,\theta) = \frac{1}{PR^2} + \frac{1}{QR^2}$, तब फलन g(b) = max. f(b, θ) $(0 \le \theta < \pi)$ जहाँ b > 0 के उन बिन्दु की संख्या होगी, जहाँ यह अवकलनीय नहीं है _________.

Ans. 1

Sol. Given that $\frac{1}{PR^2} + \frac{1}{QR^2}$ is constant

$$\frac{x-b}{\cos\theta} = \frac{y-0}{\sin\theta} = r$$

$x = b + r\cos\theta$

$y = r\sin\theta$

$r^2\sin^2\theta = 4(b + r\cos\theta)$

$r^2\sin^2\theta - 4r\cos\theta - 4b = 0$

$$r_1 + r_2 = \frac{4\cos\theta}{\sin^2\theta}$$

$$r_1 r_2 = \frac{-4b}{\sin^2\theta}$$

$$\frac{1}{PR^2} + \frac{1}{QR^2} = \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{r_1^2 + r_2^2}{r_1^2 r_2^2} = \frac{(r_1 + r_2)^2 - 2r_1 r_2}{r_1^2 r_2^2}$$

$$=\frac{\frac{16\cos^2\theta}{\sin^4\theta}+\frac{8b}{\sin^2\theta}}{\frac{16b^2}{\sin^4\theta}}=\frac{16\cos^2\theta+8b\sin^2\theta}{16b^2}$$

$$\frac{16+8b+(16-8b)\cos 2\theta}{32b^2}$$

$\therefore$ Max. value of $\frac{1}{PR^2}+\frac{1}{QR^2}=\frac{1}{b^2}$, if $b<2$

$=\frac{16b}{32b}=\frac{1}{2b}$ if $b\geq 2$. Hence, there is only one point of non differentiability

D09/4/292. If the difference between the greatest abd least values of the function $f(x)=\left(3-\sqrt{4-x^2}\right)^2+\left(1+\sqrt{4-x^2}\right)^3$ is p, then find $\left[\frac{p}{2}\right]$. (where [.] represents greatest integer function). _________.

फलन $f(x)=\left(3-\sqrt{4-x^2}\right)^2+\left(1+\sqrt{4-x^2}\right)^3$ के अधिकतम तथा न्यूनतम मानों का अंतर p है, तब $\left[\frac{p}{2}\right]$ का मान ज्ञात करो (जहाँ [.] महत्तम पूर्णांक फलन को प्रदर्शित करता है) _________.

Ans. 9

Sol. Let $t=\sqrt{4-x^2}$, $|x|\leq 2$ and $t\in[0,2]$

Let $g(t)=(3-t)^2+(1+t)^3$ for $t\in[0,2]$

$g'(t)=(t+3)(3t-1)=0$

$\therefore t=-3,\frac{1}{3}$

Also, $g''(t)=2(3t+4)$ $\qquad\therefore t=\frac{1}{3}$ is a point of local minimum at the end points

$g(0)=10$, $g(2)=28$, $g\left(\frac{1}{3}\right)=\frac{256}{27}$

$\therefore p=28-\frac{256}{27}=\frac{500}{27}$

D09/4/293. The sum of max. and the min. values of $f(x) = x^3 - 3x$ subject to condition $x^4 + 36 \le 13x^2$ _________.

फलन $f(x) = x^3 - 3x$ के अधिकतम तथा न्यूनतम मानों का योग होगा जो निम्न शर्त के अधीन हो $x^4 + 36 \le 13x^2$ _________.

Ans. 0

Sol. Given condition is

$x^4 + 36 - 13x^2 \le 0$

$x^4 - 13x^2 + 36 \le 0$

$(x^2 - 9)(x^2 - 4) \le 0$

$(x + 3)(x - 3)(x + 2)(x - 2) \le 0$

$x \in [-3, -2] \cup [2, 3]$

Now, $f(x) = x^3 - 3x$

$f'(x) = 3x^2 - 3$

For max or min $f'(x) = 0$ gives

$x^2 - 1 = 0$

$x = \pm 1$

Now, $f(1) = 1 - 3 = -2$

$f(-1) = -1 + 3 = 2$

$f(2) = 8 - 6 = 2$

$f(-2) = -8 + 6 = -2$

$f(3) = 27 - 9 = 18$

$f(-3) = -27 + 9 = -18$

Thus, the maximum value of f(x) is 18 at x = 3 and the minimum value of f(x) is –18 at x = –3.

D09/4/294. The greatest value of the function $f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1}$ _________.

फलन $f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1}$ का अधिकतम मान होगा _________.

Ans. 16

Sol. We have $f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1} = \frac{x^4 + 4x^3 + 6x^2 + 4x + 1}{x^4 - x^3 + x^2 - x + 1} = \frac{\left(x^2 + \frac{1}{x^2}\right) + 4\left(x + \frac{1}{x}\right) + 6}{\left(x^2 + \frac{1}{x^2}\right) - \left(x + \frac{1}{x}\right) + 1}$

Let $g(x) = \left(x + \frac{1}{x}\right)$, $g'(x) = \left(1 - \frac{1}{x^2}\right)$

$g'(x) = 0$ gives $\left(1 - \frac{1}{x^2}\right) = 0$

$x = \pm 1$; Hence, the greatest value of f(x) is $f(1) = \frac{2+8+6}{2-2+1} = 16$

D06/4/295. If $f(x) = a\cos(\pi x) + b$, $f'\left(\frac{1}{2}\right) = \pi$ and $\int_{\frac{1}{2}}^{\frac{3}{2}} f(x)\,dx = \frac{2}{\pi} + 1$, then find the value of $-\frac{12}{\pi}\left(\frac{\sin^{-1} a}{3} + \cos^{-1} b\right)$ _________.

यदि $f(x) = a\cos(\pi x) + b$, $f'\left(\frac{1}{2}\right) = \pi$ तथा $\int_{\frac{1}{2}}^{\frac{3}{2}} f(x)\,dx = \frac{2}{\pi} + 1$, तब $-\frac{12}{\pi}\left(\frac{\sin^{-1} a}{3} + \cos^{-1} b\right)$ का मान होगा _________.

Ans. 2

Sol. $f(x) = a\cos(\pi x) + b$

$f'(x) = -a\pi \sin(\pi x)$

$$\int_{\frac{1}{2}}^{\frac{3}{2}} f(x)\,dx = -\frac{2a}{\pi} + b = \frac{2}{\pi} + 1 \quad \Rightarrow \quad a = -1, \quad b = 1$$

D06/4/296. Let $\alpha(x) = f(x) - f(2x)$ & $\beta(x) = f(x) - f(4x)$

and $\alpha'(1) = 5$ $\alpha'(2) = 7$

then find the value of $\beta'(1) - 10$ _________.

माना $\alpha(x) = f(x) - f(2x)$ & $\beta(x) = f(x) - f(4x)$

तथा $\alpha'(1) = 5$ $\alpha'(2) = 7$

$\beta'(1) - 10$ का मान ज्ञात कीजिए _________.

Ans. 9

Sol. $\alpha'(x) = f'(x) - 2f'(2x)$ $\beta'(x) = f'(x) - 4f'(4x)$

$\alpha'(1) = f'(1) - 2f'(2) = 5$

$\alpha'(2) = f'(2) - 2f'(4) = 7$

$\beta'(1) = f'(1) - 4f'(4)$

$= \alpha'(1) + 2\alpha'(2) = 5 + (2 \times 7) = 19$

$\beta'(1) - 10 = 19 - 10 = 9$

D06/4/297. Let $f(x) = -4 \cdot e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ and g be inverse function of f and $h(x) = \frac{a + bx^{\frac{3}{2}}}{x^{\frac{5}{4}}}$, $h'(5) = 0$, then

$\frac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} = $ ________.

माना $f(x) = -4 \cdot e^{\frac{1-x}{2}} + \frac{x^3}{3} + \frac{x^2}{2} + x + 1$ तथा g, f का व्युत्क्रम फलन है तथा $h(x) = \frac{a + bx^{\frac{3}{2}}}{x^{\frac{5}{4}}}$, $h'(5) = 0$, तब

$\frac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} = $ ________.

Ans. 5

Sol.

$g(f(x)) = x$

$g'(f(x))\, f'(x) = 1$

$f(x) = -7/6$

$\therefore \quad x = 1$

$g'(-7/6)\, f'(1) = 1$

$$g'\left(-\frac{7}{6}\right) = \frac{1}{f'(1)}$$

$$f'(x) = -4 \cdot e^{\frac{1-x}{2}}\left(-\frac{1}{2}\right) + x^2 + x + 1$$

$f'(1) = 2 + 1 + 1 + 1 = 5$

$$h(x) = ax^{-\frac{5}{4}} + bx^{\frac{1}{4}}$$

$$h'(x) = -\frac{5a}{4} x^{\frac{-9}{4}} + \frac{b}{4} . x^{-\frac{3}{4}}$$

$$h'(5) = 0 \Rightarrow -\frac{5a}{4} \cdot 5^{\frac{-9}{4}} + \frac{b}{4} 5^{\frac{-3}{4}} = 0$$

$$\Rightarrow 5a \cdot 5^{\frac{-3}{2}} = b$$

$$\Rightarrow \frac{a}{b} = 5^{\frac{1}{2}}$$

$$\left(\frac{a}{b}\right)^2 = 5$$

$$\frac{a^2}{5b^2 g'\left(\frac{-7}{6}\right)} = \frac{5}{5 \times \frac{1}{5}} = 5$$

D06/4/298. Let f be a continuous function on $[0, \infty)$ such that $\lim\limits_{x \to \infty}\left(f(x) + \int_0^x f(t)\,dt\right)$ exists. Find $\lim\limits_{x \to \infty} f(x)$ ________.

माना f, $[0, \infty)$ में एक सतत् फलन इस प्रकार है कि $\lim\limits_{x \to \infty}\left(f(x) + \int_0^x f(t)\,dt\right)$ विद्यमान है। तब $\lim\limits_{x \to \infty} f(x)$ ज्ञात कीजिए ________.

Ans. 0

Sol. Let $\underset{x \to \infty}{\mathrm{Lt}}\left(f(x) + \int_0^x f(t)\,dt\right) = \ell$ ...(1)

$$\underset{x \to \infty}{\mathrm{Lt}} \frac{\left(e^x \int_0^x f(t)\,dt\right)'}{\left(e^x\right)'} = \ell$$

$$\Rightarrow \underset{x \to \infty}{\mathrm{Lt}} \frac{e^x \int_0^x f(t)\,dt}{e^x} = \ell$$

$$\Rightarrow \underset{x \to \infty}{\mathrm{Lt}} \int_0^x f(t)\,dt = \ell \qquad ...(2)$$

From (1) and (2) we get $\underset{x \to \infty}{\mathrm{Lt}} f(x) = 0$

D06/4/299. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and let $g(x) = f^{-1}(x)$. Find $g'''(0)$. ________.

माना $f(x)=x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\frac{x^5}{5}$ तथा $g(x) = f^{-1}(x)$ है। $g'''(0)$ ज्ञात कीजिए _________.

Ans. 1

Sol. $f(0) = 0, f'(0) = 1, f''(0) = 1\ f'''(0) = 2$

$g(f(x))=x \quad \Rightarrow g'(f(x))\, f'(x)=1$

$$\Rightarrow g''(f(x))=\frac{-f''(x)}{(f'(x))^3}$$

$$\Rightarrow g'''(f(x))f'(x)=-\left[\frac{(f'(x))^3.f'''(x)-3(f''(x))^2(f'(x))^2}{(f'(x))^6}\right]$$

Put $x = 0$

$$g'''(0)1=-\left[\frac{1\times 2-3\times 1}{1}\right]=1$$

D08/4/300. Let $f(x) = x^3 - 6x^2 + ax - 8$ be a function such that $f(x) = 0$ has three positive real roots. The function $y = |f(|x|)|$ has 'b' critical points then the value of $|a - b|$ equals _________.

माना $f(x) = x^3 - 6x^2 + ax - 8$ एक फलन इस प्रकार है कि $f(x) = 0$ के तीन धनात्मक वास्तविक मूल हैं। फलन $y = |f(|x|)|$ के क्रांतिक बिन्दुओं की संख्या 'b' हैं तो $|a - b|$ का मान बराबर होगा _________.

Ans. 9

Sol. $x^3 - 6x^2 + ax - 8 = 0$ has only +ve roots

$\Rightarrow \alpha = \beta = \gamma = 2$

$\Rightarrow a = 12 \Rightarrow f(x) = (x - 2)^3$

$\Rightarrow y = |f(|x|)|$ has three critical points

$\Rightarrow b = 3$

D08/4/301. The minimum value of the real constant 'a' can be expressed as $\frac{p}{q}$ where p and q are relatively prime positive integer for which $f'(x) \ge 0\ \forall\ x \in R$ for the function $f(x)=ax-\frac{x^3}{1+x^2}$. Find $(p - q)$ _________.

एक वास्तविक अचर 'a' के न्यूनतम मान को $\frac{p}{q}$ के रूप में लिखा जाता है जहाँ p तथा q परस्पर अभाज्य संख्याएँ हैं जिनके लिए

फलन $f(x) = ax - \frac{x^3}{1+x^2}$ के लिए $f'(x) \geq 0 \ \forall \ x \in R$ है। (p – q) का मान ज्ञात कीजिए _________.

Ans. 1

Sol. $f'(x) = a - \frac{3x^2+x^4}{(1+x^2)^2} \geq 0$

$a \geq \frac{3x^2+x^4}{1+2x^2+x^4} \quad \forall \ x \in R$

Let $x^2 = t$

$a \geq \underbrace{\frac{3t+t^2}{1+2t+t^2}}_{\text{Range:}\left[0,\frac{9}{8}\right]} \quad \forall \ t \in [0, \infty)$

$\therefore \quad a \geq \frac{9}{8}$

D08/4/302. Let y = f(x) be a differentiable function such that f(1) = 1, f(2) = –3, f(3) = 4, f(4) = –1 and f(5) = 2. Let y = g(x) be a differentiable function such that $f'(x)\,g(x) \neq g'(x)\,f(x)$ then minimum number of real roots of the equation $(g'(x))^2 + g(x)g''(x) = 0$ is _________.

माना y = f(x) एक अवकलनीय फलन इस प्रकार है कि f(1) = 1, f(2) = –3, f(3) = 4, f(4) = –1 तथा f(5) = 2 हैं। माना y = g(x) एक अवकलनीय फलन इस प्रकार है कि $f'(x)\,g(x) \neq g'(x)\,f(x)$ तो समीकरण $(g'(x))^2 + g(x)g''(x) = 0$ के वास्तविक मूलों की न्यूनतम संख्या होगी _________.

Ans. 4

Sol. f(x) = 0 has at least 4 real roots

$f'(x) \,.\, g(x) \neq g'(x)f(x) \Rightarrow g(x) = 0$ has at least 3 real roots

$\Rightarrow g'(x) \,.\, g(x) = 0$ has at least 5 real roots

$\Rightarrow (g'(x))^2 + g(x)\,g''(x) = 0$ has at least 4 real roots

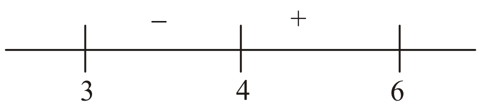
D08/4/303. Let f(x) be a function defined by $f(x) = \int_3^x t\left(t^2 - 7t + 12\right)dt$, $3 \leq x \leq 6$. If the range of f(x) is [a, b], then the value of $\left(3a + \frac{b}{9}\right)$ is _________.

माना f(x) एक फलन इस प्रकार परिभाषित है कि $f(x)=\int_3^x t\left(t^2-7t+12\right)dt$, $3 \le x \le 6$ है। यदि f(x) का परिसर [a, b] है, तो $\left(3a+\frac{b}{9}\right)$ का मान होगा _________.

Ans. 1

Sol. $f(x)=\int_3^x t\left(t^2-7t+12\right)dt$

$f'(x) = x(x-3)(x-4)$



minimum at $x = 4$

maximum at $x = 3$ or 6

$\therefore f(3) = 0,\ f(6)=\int_3^6\left(t^3-7t^2-12t\right)dt$

$=\left(\frac{t^4}{4}-\frac{7t^3}{3}+\frac{12t^2}{2}\right)_3^6=\frac{99}{4}$

$f(x)=\int_3^4\left(t^3-7t^2+12t\right)dt=-\frac{7}{12}$

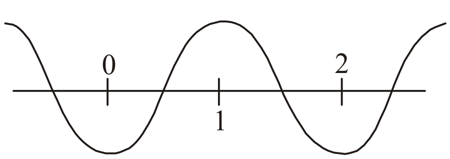
$\therefore 3a+\frac{b}{9}=1$

D08/4/304. The number of distinct real roots of equation $2x^4 - 8x^3 + 8x^2 - 1 = 0$ is n, then the value of $\frac{1}{n}$ is _________.

समीकरण $2x^4 - 8x^3 + 8x^2 - 1 = 0$ के भिन्न वास्तविक मूलों की संख्या n है, तो $\frac{1}{n}$ का मान होगा _________.

Ans. 0.25

Sol.



$f(x) = 2x^4 - 8x^3 + 8x^2 - 1$

$f'(x) = 8x^3 - 24x^2 + 16x$
$= 8x(x^2 - 3x + 2)$
$= 8x(x - 1)(x - 2)$
$f(0) = -1, f(1) = 1, f(2) = -1$
So, f(x) has 4 distinct roots.

D08/4/305. Let f be a continuous function and satisfies $f'(x) > 0$ on $(0, \infty)$ and the value of $f''(x)\ \forall\ x \in (0, \infty)$ is equal to minimum value of $\min\limits_{x\in R}\left\{e^{-|x|} + 2, |x| + 2\right\}$.

If $\lim\limits_{x\in R} \dfrac{3x^2 - \dfrac{3}{x^2+1} - 4f'(x)}{f(x)} = L$ then find the value of $[L^2]$. [Note : [k] denotes greatest integer less than or equal to k]

माना f एक सतत् फलन है तथा $f'(x) > 0$ को अंतराल $(0, \infty)$ में संतुष्ट करता है तथा $f''(x)\ \forall\ x \in (0, \infty)$ का मान $\min\limits_{x\in R}\left\{e^{-|x|} + 2, |x| + 2\right\}$ के न्यूनतम मान के बराबर है।

यदि $\lim\limits_{x\in R} \dfrac{3x^2 - \dfrac{3}{x^2+1} - 4f'(x)}{f(x)} = L$ है तो $[L^2]$ का मान ज्ञात कीजिए। [नोट : [k], k के बराबर या उससे कम महत्तम पूर्णांक को दर्शाता है]

Ans. 9

Sol. $\forall\ x \in R : |x| + 2 \in [2, \infty)$
$e^{-|x|} + 2 \in (2, 3]$
$\Rightarrow \quad \forall\ x \in (0, \infty) : f''(x) = 2$
$\Rightarrow \quad \forall\ x \in (0, \infty) : f'(x) = 2x + c$ for some constant c.

$f'(x) > 0\ \forall\ x \in (0, \infty) \Rightarrow c > 0$ because $\lim\limits_{x\to 0^+} f'(x) = c$ for some constant d.

Also $f(x) = x^2 + cx + d$ for some constant d.

$$L = \lim_{x\to\infty} \frac{3x^2 - \dfrac{3}{x^2+1} - 4(2x+c)}{x^2 + cx + d} = 3$$

D08/4/306. Let f(x) be a continuous and differentiable function satisfying the following conditions

(a) $\prod_{r=1}^{6} f(r) < 0, \prod_{r=1}^{3} f(2r) < 0, \prod_{r=1}^{2} f(3r) < 0$

(b) $f(6) > 0$, and

(c) $f(x)$ is monotonic in $(n, n+1), n \in I$

Let A denotes the set consisting of number of distinct possible roots of $f(x) = 0$ in $x \in (1, 6)$. Find the sum of all the elements of set A.

माना कि f(x) एक सतत् तथा अवकलनीय फजन है जो निम्न शर्तों को संतुष्ट करता है :

(a) $\prod_{r=1}^{6} f(r) < 0, \prod_{r=1}^{3} f(2r) < 0, \prod_{r=1}^{2} f(3r) < 0$

(b) $f(6) > 0$, तथा

(c) $f(x), (n, n+1), n \in I$ में एकदिष्ट है

माना A एक समुच्चय जो अंतराल (1, 6) में $f(x) = 0$ के विभिन्न मूलों से निर्मित है, तब समुच्चय A के अवयवों का योग ज्ञात कीजिए।

Ans. 10

Sol. Since $f(6) > 0$

and $\prod_{r=1}^{2} f(3r) < 0$

$f(3) < 0$

Also $\prod_{r=1}^{3} f(2r) < 0$

$\Rightarrow f(2) \times f(4) \times f(6) < 0$

$\Rightarrow f(2) \times f(4) < 0$ ...(i)

and $\prod_{r=1}^{6} f(r) < 0$

$\Rightarrow (f(1) \times f(5)) \times \underset{-ve}{\xleftrightarrow{(f(3)\times f(4))}} \times \underset{-ve}{\xleftrightarrow{(f(3)\times f(6))}} < 0$

$\Rightarrow f(1) \times f(5) < 0$ ...(ii)

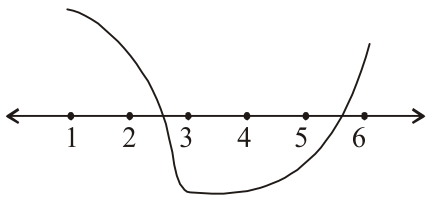
Hence, f(1) and f(5) have opposite signs.

and f(2) and f(4) have opposite signs

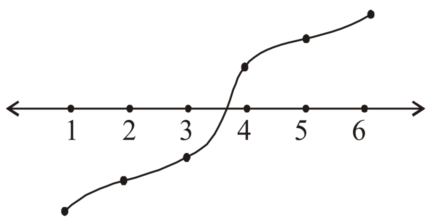
4 cases arise w.r.t. signs of f(1) and f(2)

Case 1 : f(1) and f(2) > 0

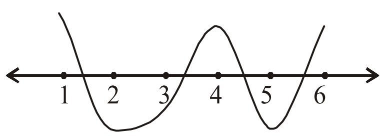
$\Rightarrow$ f(5) and f(4) < 0



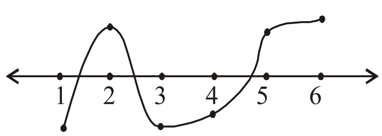
Graphically or otherwise :
2 distinct roots of f(x) = 0
1 in (2,3) and 1 in (5,6)
Case 2 : f(1) and f(2) < 0
$\Rightarrow$ f(5) and f(4) > 0]



1 distinct root of f(x) = 0 in (3,4)
Case 3 : f(1) > 0 and f(2) < 0
$\Rightarrow$ f(5) <0 and f(4) > 0



4 distinct roots of f(X) = 0 in 1 each in : (,12), (3,4), (4,5) and (5,6)
Case 4 : f(1) < 0 and f(2) > 0
$\Rightarrow$ f(5) > 0 and f(4) < 0



$\Rightarrow$ 3 distinct roots of f(x) = 0
1 each in (1,2), (2,3) and (4,5)
Hence, A = {1,2,3,4} from the 4 possible cases.

D08/4/307. Let $f(x) = \frac{\pi}{2} + \tan^{-1} x$, where sgn(y) denotes signum function of y, and g(x) is the inverse of f(x). If the complete set of values of k for which the equation $2g(x) + k(\pi - 2x) = 0$ has three distinct solutions is (a, ∞) then find 'a' _________.

माना $f(x)=\frac{\pi}{2}+\tan^{-1}x$, जहाँ sgn(y), y के चिह्न फलन को प्रदर्शित करता है तथा g(x), f(x) का प्रतिलोम है। यदि k के मानों का पूर्ण समुच्चय जिसके लिए समीकरण $2g(x)+k(\pi-2x)=0$ के तीन विभिन्न हल हों, (a, ∞) है तब 'a' का मान ज्ञात कीजिए।

Ans. 1

Sol. $|\text{sgn}(f(x))| = 1 \quad \forall x : f(x) \neq 0$

$0 \quad \forall x : f(x) = 0$

$$\Rightarrow \left|\text{sgn}\left(\tan^{-1}\frac{x}{1+x^2}\right)\right| = \begin{Bmatrix} 1 & \forall x \in R-\{0\} \\ 0 & x=0 \end{Bmatrix}$$

$$\Rightarrow f(x)=\frac{\pi}{2}+\tan^{-1}x \forall x \in R$$

$f : R \to (0, \pi) \Rightarrow g = f^{-1} : (0, \pi) \to R$

Also $\quad g(x)=\tan\left(x-\frac{\pi}{2}\right)$

$2g(x) + k(\pi - 2x) = 0$

$$\Rightarrow 2k\left(\frac{\pi}{2}-x\right)=2\tan\left(\frac{\pi}{2}-x\right)$$

Let $\quad y=\frac{\pi}{2}-x : y \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

and we've to find solutions of $\tan y = ky$

Note that : $\tan \alpha = k\alpha \Leftrightarrow \tan(-\alpha) = k(-\alpha)$

$y = 0$ is a root and all non-zero roots occur in pairs.

Hence, this equation will've 3 roots in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ iff there's a solution in $\left(0,\frac{\pi}{2}\right)$.

Consider : $h(y) = \tan y - ky$

At $y = 0$, $h(0) = 0$

As $y \to \frac{\pi}{2}$, $h(y) \to \infty$

Also $h'(y) = \sec^2 y - k$

$\forall\ k \le 1$, $h'(y) \ge 0 \Rightarrow h(y)$ increases in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$$\Rightarrow h(y) > 0 \forall y \in \left(0,\frac{\pi}{2}\right)$$

No solution for $h(y) = 0$ in $\left(0, \frac{\pi}{2}\right)$.

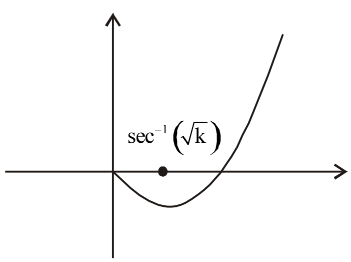
Let $k > 1$

$\Rightarrow h'(0) = 1 - k < 0$

$\Rightarrow h(y)$ decreases as y starts increasing from O until $y = \sec^{-1}\left(\sqrt{k}\right)$ and after that h(y) increases to infinity as $y \to \frac{\pi}{2}$

Graphically, or otherwise, when $k \in (1, \infty)$, $h(y) = 0$ has exactly 1 solution in $\left(0, \frac{\pi}{2}\right)$ and 3 solutions in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Graph of h(y) for $k > 1$ in $\left(0, \frac{\pi}{2}\right)$



D08/4/308. Let f be a twice differentiable function defined in [–3,3] such that $f(0) = 0$, $f'(3) = 0$, $f'(-3) = 12$ and $f''(x) \geq -2 \ \forall \ x \in [-3,3]$. If $g(x) = \int_0^x f(t)\,dt$ then find maximum value of g(x).

यदि अंतराल [–3,3] में परिभाषित f एक द्विअवकलनीय फलन इस प्रकार है कि $f(0) = 0$, $f'(3) = 0$, $f'(-3) = 12$ तथा $f''(x) \geq -2 \ \forall \ x \in [-3,3]$ हैं। यदि $g(x) = \int_0^x f(t)\,dt$ है तो g(x) का अधिकतम मान ज्ञात कीजिए।

Ans. 48

Sol. $\forall \ x \in [-3,3] : f''(x) \geq -2$

$\Rightarrow f''(x) + 2 \geq 0$

$\Rightarrow \frac{d}{dx}\left(f'(x)+2x\right) \geq 0$

$\Rightarrow f'(x) + 2x$ is non-decreasing in $[-3,3]$

$f'(3) = 0$ and $f'(-3) = 12$

$\Rightarrow f'(-3) - 2(-3) = f'(3) - 2 \times 3 = 6$

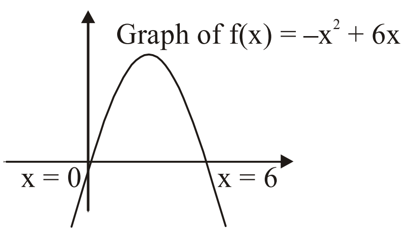
$\Rightarrow f'(x) + 2x$ is a constant function $\forall\ x \in [-3,3]$

$\Rightarrow f'(x) = -2x + 6$

$\Rightarrow f(x) = -x^2 + 6x$

Graphically or otherwise

$g(x) = \int_0^x f(t)\,dt$ attains its maximum value at $x = 6$



Maximum value of $g(x) = \int_0^6 \left(-x^2 + 6x\right)dx$

$= \int_{-3}^{3} \left(9 - t^2\right)dt$ {Substitute $t = x - 3$}

$= 2\int_0^3 \left(9 - t^2\right)dt = 2[27 - 9] = 36$

D08/4/309. Let the function $f(x) = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g (a \neq 0)$ touches the line $y = g(x) = x + p$ at $x = 1$, 2 and 3 and $h(x) = f(x) - g(x)$. If $\int_1^3 h(x)\,dx = \frac{32}{105}$, find the value of a.

माना फलन $f(x) = ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g (a \neq 0)$ रेखा $y = g(x) = x + p$ को $x = 1, 2$ तथा 3 पर स्पर्श करता है तथा $h(x) = f(x) - g(x)$ है। यदि $\int_1^3 h(x)\,dx = \frac{32}{105}$ है, तो a का मान ज्ञात कीजिए।

Ans. 2

Sol.

D08/4/310. Number of non-similar isosceles triangle ABC such that $\tan A + \tan B + \tan C = 100$.

असमान समद्विबाहु त्रिभुज ABC की संख्या होगी यदि $\tan A + \tan B + \tan C = 100$ है :

Ans. 2

Sol. Let A = B, then $2A + C = 180^\circ$ and $2 \tan A + \tan C = 100$

Now $2A + C = 180^\circ \Rightarrow \tan 2A = -\tan C$ ...(i)

Also $2 \tan A + \tan C = 100$

$2 \tan A - 100 = -\tan C$ ...(ii)

From (i) and (ii) $2 \tan A - 100 = \dfrac{2 \tan A}{1 - \tan^2 A}$

Let $\tan A = x$, then $\dfrac{2x}{1-x^2} = 2x - 100$

$x^3 - 50x^2 + 50 = 0$

Let $f(x) = x^3 - 50x^2 + 50$

Then $f'(x) = 3x^2 - 100x$. Thus $f'(x) = 0$ has roots 0, $\dfrac{100}{3}$. Also $f(0) > 0$, $f\left(\dfrac{100}{3}\right) < 0$. Thus $f(x) = 0$ has exactly three distinct real roots. Therefore tan A and hence A has three distinct values, but one of them will be obtuse angle. Hence there exists exactly two non similar isosceles triangles.

D08/4/311. $x, y \in R$, $x^2 + y^2 + xy = 1$. Find the maximum value of $|x^2y + xy^3|$.

$x, y \in R$, $x^2 + y^2 + xy = 1$ है। $|x^2y + xy^3|$ का अधिकतम मान ज्ञात कीजिए।

Ans. 2

Sol. Put $x = r \cos\theta$, $y = r \sin\theta$

We get $r^2 + r^2 \sin\theta \cos\theta = 1$

$\Rightarrow r^2 = \dfrac{2}{2 + \sin 2\theta} \Rightarrow \dfrac{2}{3} \le r^2 \le 2$ and $-1 \le \left(1 - r^2\right) \le \dfrac{1}{3}$

Let $F = x^3y + xy^3 = (1 - r^2)\, r^2$

$F = (1 - t)t$ where $t = r^2$, hence $\dfrac{2}{3} \le t \le 2$

$\dfrac{dF}{dt} = 1 - 2t < 0$ for $t \in \left[\dfrac{2}{3}, 2\right]$

Hence F is decreasing

$\therefore F\left(\dfrac{2}{3}\right) = \dfrac{2}{9}$ and $F(2) = -2$

$\therefore F \in \left[-2, \frac{2}{9}\right]$ $\therefore$ $0 \le |F| \le 2$

D08/4/312. Let $f(x) = \left\{b^2 + (a-1)b - \frac{1}{4}\right\}x + \int_0^x \left(\sin^4 x + \cos^4 x\right)dx$. If f(x) be a non-decreasing function of x, $\forall x \in$ R and $\forall b \in$ R. Then the number of positive integer values of parameter a is _________.

माना $f(x) = \left\{b^2 + (a-1)b - \frac{1}{4}\right\}x + \int_0^x \left(\sin^4 x + \cos^4 x\right)dx$ है। यदि फलन f(x) $\forall x \in$ R तथा $\forall b \in$ R के लिए ऐसा फलन है जो ह्रासमान नहीं है। तब प्राचल a के धनात्मक पूर्णांकों की संख्या होगी _________.

Ans. 2

Sol. $f'(x) = \left\{b^2 + (a-1)b - \frac{1}{4}\right\} + \left(\sin^4 x + \cos^4 x\right)$

Since function is non-decreasing

$f'(x) \ge 0$ $\forall$ x $\in$ R and $\forall$ b $\in$ R

$\Rightarrow$ $\sin^4 x + \cos^4 x \ge -b^2 - (a-1)b + \frac{1}{4}$

$\Rightarrow$ $-b^2 - (a-1)b + \frac{1}{4} \le \left(\sin^4 x + \cos^4 x\right)$ ...(i)

But minimum value of $\sin^4 x + \cos^4 x$ is $\frac{1}{2}$

Thus from equation (i)

$-b^2 - (a-1)b + \frac{1}{4} \le \frac{1}{2}$ $\forall$ b $\in$ R

$b^2 + (a-1)b + \frac{1}{4} \ge 0$ $\forall$ b $\in$ R

Now, $D \le 0$

$\Rightarrow (a-1)^2 - 4(1)\left(\frac{1}{4}\right) \le 0$

$\Rightarrow$ $(a-1)^2 - 1 \le 0$ $\Rightarrow$ $a \in [0,2]$

D08/4/313. $f(x)$ and $g(x)$ are polynomials of degree 3. $g(\alpha) = g'(\alpha) = 0$. If $\lim\limits_{x\to\alpha} \frac{f(x)}{g(x)} = 0$, then find the number of different solutions of equation $f(x)g'(x) + f'(x)g(x) = 0$ _________.

$f(x)$ तथा $g(x)$ त्रिघातीय बहुपद हैं। $g(\alpha) = g'(\alpha) = 0$ है। यदि $\lim\limits_{x\to\alpha} \frac{f(x)}{g(x)} = 0$ है, तो समीकरण $f(x)g'(x) + f'(x)g(x)$ $= 0$ के भिन्न हलों की संख्या ज्ञात कीजिए _________.

Ans. 2

Sol. For existence of the given limit $f(\alpha) = f'(\alpha) = f''(\alpha) = 0$

$\because$ $f(x)$ is polynomial of degree 3 $\Rightarrow$ $f(x) = \lambda(x - \alpha)^3$

Similarly $g(x) = \mu(x - \alpha)^2 (x - \beta)$ $(\beta \neq \alpha)$

$\Rightarrow$ $f(x)g(x) = \lambda\mu(x - \alpha)^5 (x - \beta)$

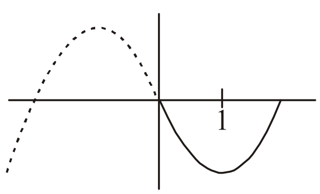
$\therefore$ Number of different solution of $\frac{d}{dx}\left(f(x).g(x)\right) = 0$ are 2 i.e. one solution α and one between α and β.

D08/4/314. If the equation $x^3 - 3x + |3x - a| \le 0$ is to be satisfied by atleast one $x > 0$, then let M be the interval of values of a needed, then integral part of length of M is _________.

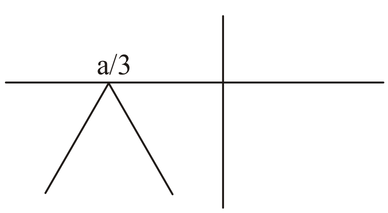
यदि समीकरण $x^3 - 3x + |3x - a| \le 0$ को कम से कम एक $x > 0$ द्वारा संतुष्ट किया जाता है, तो माना M, a के आवश्यक मानों का अंतराल है, तो M की लम्बाई का पूर्णांक भाग होगा _________.

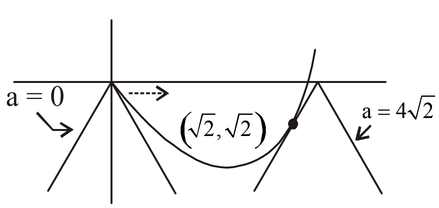
Ans. 5

Sol.



$-|3x - a|$ is an inverted 'V' shape





$x^3 - 3x \le -|3x - a|$

Now for the above condition to hold true.

$$\frac{d}{dx}\left(x^3-3x\right)_{(x_1,y_1)}=3x_1^3-3$$

$3x_1^3-3=\pm3,\ x_1>0$

$x_1=0,\ x_1=\sqrt{2}$

$y_1=0,\ y_1=\left(\sqrt{2}\right)^3-3\sqrt{2}=-\sqrt{2}$

The points of intersection of $-|3x-a|$ with x^3-3x as shown in figure above is (0,0) and $\left(\sqrt{2},-\sqrt{2}\right)$

For point of intersection to be (0,0), a = 0

For point of intersection to be $\left(\sqrt{2},-\sqrt{2}\right)$

$a=4\sqrt{2}$

$0\le a\le 4\sqrt{2}$

$\left[0,4\sqrt{2}\right]$ is M

$[\text{Length}]=\left[4\sqrt{2}\right]=5$

D08/4/315. Let f(x) be a non-constant thrice differentiable function defined on $(-\infty,\infty)$ such that $f(x)=f(6-x)$ and $f'(0)=0=f'(2)=f'(5)$. If n is the minimum number of roots of $(f''(x)^2)+f'(x)f'''(x)=0$ in the interval [0,6], then the value of n is ________.

माना f(x) अंतराल $(-\infty,\infty)$ में एक अचर तीन बार अवकलनीय फलन इस प्रकार है कि $f(x)=f(6-x)$ तथा $f'(0)=0=f'(2)=f'(5)$ हैं। यदि $(f''(x)^2)+f'(x)f'''(x)=0$ के अंतराल [0,6] में मूलों की न्यूनतम संख्या n है, तो n का मान होगा ________.

Ans. 12

Sol. On differentiating (1) w.r.t. x, we get

$f'(x)=-f'(6-x)$

Putting x = 0,2,3,5 in (2), we get

$f'(0)=-f'(6)=0$

Similarly, $f'(2)=f'(4)=0$

$f'(3)=0$

$f'(5)=f'(1)=0$

$\therefore\ f'(0)=0=f'(2)=f'(3)=f'(5)=f'(1)=f'(4)=f'(6)$

Therefore, $f'(x)=0$ has minimum seven roots in [0,6]

Now, consider a function $y=f'(x)$.

As $f'(x)$ satisfies Rolle's theorem in intervals [0,1], [1,2], [2,3], [3,4], [4,5], and [5,6], respectively, by Rolle's theorem, the equation $f''(x) = 0$ has minimum six roots.

Now, $g(x) = (f''(x))^2 + f'(x)\,f'''(x) = 0 = h'(x)$, where

$h(x) = f'(x)\,f''(x)$.

Clearly, $h(x) = 0$ has minimum 13 roots in [0,6].

D07/4/316. Let $f : [0, \infty) \to R$ be a continuous, strictly increasing function such that $f^3(x) = \int_0^x t\,f^2(t)dt$. If a normal drawn to curve $y = f(x)$ with gradient $-\frac{1}{2}$, then find the intercept made by it on y-axis _________.

माना $f : [0, \infty) \to R$ संतत्, निरन्तर वर्धमान फलन इस प्रकार है कि $f^3(x) = \int_0^x t\,f^2(t)dt$. यदि वक्र $y = f(x)$ पर $-\frac{1}{2}$ का अभिलम्ब खींचा जाता है, तो इसके द्वारा y-अक्ष पर काटा गया अन्तःखण्ड होगा _________.

Ans. 9

Sol. $3f^2(x).\,f'(x) = x\,.\,f^2(x)$

$\Rightarrow f'(x)\frac{x}{3} \qquad (\because\ f^2(x) \geq 0)$

$f(x) = \frac{x^2}{6} \qquad (\because\ f(0) = 0)$

Equation of normal at (x_1, y_1)

$y - y_1 = -\frac{3}{x_1}(x - x_1)$ | $-\frac{3}{x_1} = -\frac{1}{2}$

y-intercept $= 3 + y_1$. $\qquad x_1 = 6;\ \ y_1 = 6$

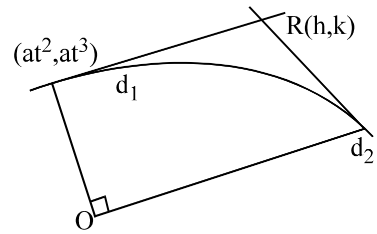
D07/4/317. A curve is defined parametrically by the equations $x = t^2$ and $y = t^3$. A variable pair of perpendicular lines through the origin 'O' meet the curve at P and Q. If the locus of the point of intersection of the tangents at P and Q is $ay^2 = bx - 1$, then find the value of $(a + b)$ _________.

एक वक्र प्राचलिक रूप में समीकरणों $x = t^2$ तथा $y = t^3$ द्वारा प्रदर्शित किया जाता है। लम्बवत् रेखाओं का एक चर युग्म मूल बिन्दु 'O' से गुजरने वाले वक्र को P तथा Q पर मिलता है। यदि बिन्दु P तथा Q पर खींची गई स्पर्श रेखाओं के प्रतिच्छेद बिन्दुओं का बिन्दुपथ $ay^2 = bx - 1$ है, तब $(a + b)$ का मान होगा _________.

Ans. 7

Sol. $x = at^2\ ;\ y = at^3$

$$\frac{dx}{dt} = 2at\ ;\ \frac{dy}{dt} = 3at^2$$



$$\frac{dy}{dx} = \frac{3t}{2}$$

$$y - at^3 = \frac{3t}{2}(x - at^2) \quad ....(1)$$

$$2k - 2at^3 = 3th - 3at^3$$
$$at^3 - 3th + 2k = 0$$

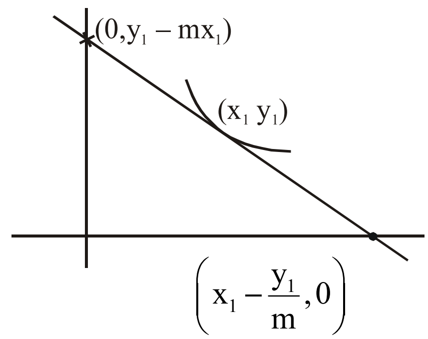
$$t_1t_2t_3 = -\frac{2k}{a} \quad (\text{put } t_1t_2 = -1); \qquad \text{hence } t_3 = \frac{2k}{a}$$

now t_3 must satisfy the equation (1) which gives the required locus.

D07/4/318. Find the value of $|a|$ for which the area of triangle included between the coordinate axes and any tangent to the curve $x^a y = \lambda^a$ is constant (where λ is constant) _________.

$|a|$ का मान ज्ञात कीजिए जिसके लिए वक्र $x^a y = \lambda^a$ की किसी स्पर्श रेखा तथा निर्देशांक अक्षों द्वारा निर्मित त्रिभुज का क्षेत्रफल नियत है। (जहाँ λ एक अचर है) _________.

Ans. 1



Sol. $x^a . y = \lambda^a \quad ; \quad y = \frac{\lambda^a}{x^a}$

$$\frac{dy}{dx} = -a\lambda^a x^{-a-1} = -a\frac{x^a.y}{x^{a+1}}$$

$$\Rightarrow m = \frac{-ay_1}{x_1}s$$

$$A = \frac{1}{2}|y_1 - mx_1|\left|x_1 - \frac{y_1}{m}\right| = \frac{1}{2}y_1x_1(1+a)^2 = \frac{1}{2}\lambda^a.x_1^{1-a}(1+a)^2$$

For A to be constant $1 - a = 0$.

D07/4/319. If m is the slope of a line which is tangent to $y^3 = x^4$ & a normal to $x^2 - 2x + y^2 = 0$, then $\left(\frac{3m}{4}\right)^3$ is equal to $(m \neq 0)$ _________.

यदि m एक रेखा की प्रवणता है जो $y^3 = x^4$ पर एक स्पर्शरेखा है तथा $x^2 - 2x + y^2 = 0$ पर एक अभिलम्ब है, तब $\left(\frac{3m}{4}\right)^3$ का

मान बराबर होगा $(m \neq 0)$ _________.

Ans. 4

Sol. Let a point on $y^3 = x^4$ be (t^3, t^4)

$3y^2y' = 4x^3$

$\Rightarrow y' = \dfrac{4x^3}{3y^2}$ $\qquad\qquad\qquad \Rightarrow y' = \dfrac{4}{3}t$

Equation of tangent is

$$y - t^4 = \frac{4t}{3}\left(x - t^3\right)$$

$\because$ it is a normal to $x^2 + y^2 - 2x = 0$

$\therefore$ it must pass through $(1, 0)$

$$\Rightarrow -\frac{3}{4}t^3 = 1 - t^3 \Rightarrow \frac{t^3}{4} = 1$$

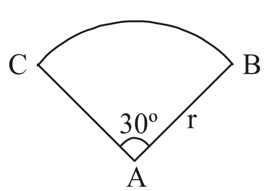
$\Rightarrow t^3 = 4$

Now $m = \dfrac{4t}{3}$

$$\Rightarrow \left(\frac{3m}{4}\right)^3 = t^3 = 4$$

D07/4/320. Consider a circular arc ABC as shown in figure, having variable radius r. The circle passing through points A, B, C is having radius R. The value of $\left(2\cos 15^\circ \times \dfrac{dR}{dr}\right)$ is equal to _________.

चित्र में दर्शाए गए चाप ABC पर विचार करें जिसकी चर त्रिज्या r है। बिन्दुओं A, B, C से गुजरने वाले वृत्त की त्रिज्या R है। तो $\left(2\cos 15^\circ \times \dfrac{dR}{dr}\right)$ का मान बराबर होगा _________.



Ans. 1

Sol. $R = \dfrac{abc}{4\Delta}$

$$= \frac{r.r.\left(2r\sin\frac{\theta}{2}\right)}{4\left(\frac{1}{2}r^2\sin\theta\right)} = \frac{r\sin\frac{\theta}{2}}{\sin\theta} = \frac{r}{2\cos\frac{\theta}{2}} = \frac{r}{2\cos 15^o}$$

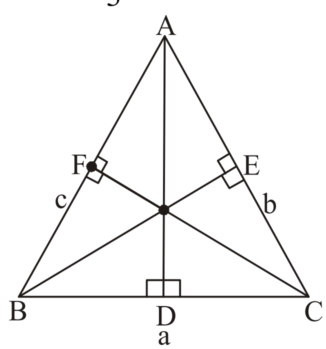
$$\therefore \frac{dR}{dr}\times 2\cos 15^o = 1$$

T03/4/321. Let the lengths of the altitudes drawn from the vertices of a ΔABC to the opposite sides are 2, 2 and 3. If the area of ΔABC is Δ, then find the value of $2\sqrt{2}\Delta$ _________.

यदि ΔABC के शीर्षो से सम्मुख भुजाओं पर डाले गये शीर्ष लम्बो के लम्बाईयाँ 2, 2 तथा 3 है। यदि ΔABC का क्षेत्रफल Δ है, तो $2\sqrt{2}\Delta$ का मान _________ होगा।

Ans. 9

Sol. We have $a = \frac{2\Delta}{h_A}; b = \frac{2\Delta}{h_B}; c = \frac{2\Delta}{h_C}$ i.e. $a = \Delta; b = \Delta; c = \frac{2\Delta}{3}$



$$\Rightarrow \quad a+b+c = 2\Delta\left(\frac{1}{h_A}+\frac{1}{h_B}+\frac{1}{h_C}\right)$$

$$\Rightarrow \quad 2s = 2\Delta\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{3}\right) = 2\Delta\left(\frac{8}{6}\right)$$

$$\Rightarrow \quad s = \frac{4}{3}\Delta \qquad \text{\_\_\_\_\_\_\_\_\_\_(1)}$$

Now $\Delta^2 = s(s-a)(s-b)(s-c)$

$$\therefore \Delta^2 = \frac{4}{3}\Delta\left(\frac{\Delta}{3}\right)\left(\frac{\Delta}{3}\right)\left(\frac{2\Delta}{3}\right) \Rightarrow \quad 1 = \frac{8}{81}\Delta^2$$

$\Rightarrow 8\Delta^2 = 81$

Hence $\sqrt{8\Delta^2} = 9$

MATCH THE COLUMN TYPE QUESTION(S)

D05/5/322. Match the following column :

Column-I	**Column-II**
(a) The number of the values of x in $(0, 2\pi)$, where the function $f(x) = \frac{\tan x + \cot x}{2} - \left\|\frac{\tan x - \cot x}{2}\right\|$ is continuous but not	(p) 2

differentiable is

(b) The number of points where the function $f(x) = \min\{1, 1+x^3, x^2-3x+3\}$ is non-derivable — (q) 0

(c) The number of points where $f(x) = (x+4)^{\frac{1}{3}}$ is non-differentiable is — (r) 4

(d) Consider $f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{x.2}{\pi}\right)+\frac{\pi}{2} & 0 < x \le \frac{\pi}{2} \\ \sin^{-1}\sin x & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$ — (s) 1

Number of points in $\left(0, \frac{3\pi}{2}\right)$, where f(x) is non-differentiable is

निम्न कॉलम को सुमेलित कीजिए :

स्तम्भ-I **स्तम्भ-II**

(a) $(0, 2\pi)$ में x के मानों की संख्या जहाँ फलन $f(x) = \frac{\tan x + \cot x}{2} - \left|\frac{\tan x - \cot x}{2}\right|$ सतत् है लेकिन अवकलनीय नहीं — (p) 2

(b) उन बिन्दुओं की संख्या जहाँ फलन $f(x) = \min\{1, 1+x^3, x^2-3x+3\}$ अवकलनीय नहीं है — (q) 0

(c) बिन्दुओं की संख्या जहाँ $f(x) = (x+4)^{\frac{1}{3}}$ अवकलनीय नहीं है — (r) 4

(d) विचार कीजिए $f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{x.2}{\pi}\right)+\frac{\pi}{2} & 0 < x \le \frac{\pi}{2} \\ \sin^{-1}\sin x & \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$ — (s) 1

$\left(0, \frac{3\pi}{2}\right)$ में बिन्दुओं की संख्या, जहाँ f(x) अवकलनीय नहीं है

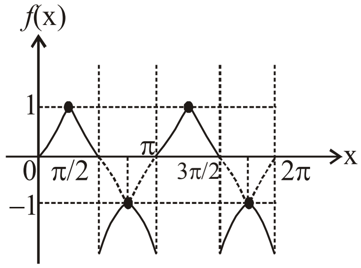
(A) $A \to p, B \to q, C \to r, D \to s$
(B) $A \to r, B \to p, C \to s, D \to q$
(C) $A \to s, B \to q, C \to r, D \to p$
(D) $A \to r, B \to q, C \to p, D \to s$

Ans. B

Sol. (A) $f(x) = \frac{\tan x + \cot x}{2} - \left|\frac{\tan x - \cot x}{2}\right|$

$$f(x)=\begin{cases}\cot x & \tan x \ge \cot x\\ \tan x & \tan x < \cot x\end{cases}$$

There are 4 points where the function is continuous but not differentiable in $(0, 2\pi)$

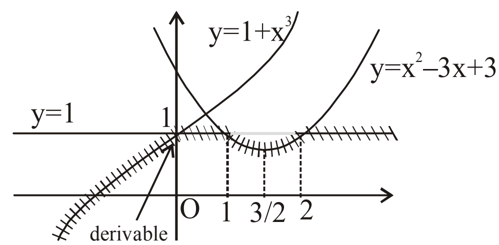


(B) This hatched graph is for f(x).

$\therefore$ f(x) is non-derivable at two points.

Ans = option (P)

(C) $f(x)=(x+4)^{\frac{1}{3}}$



$$f'(x)=\frac{1}{3}(x+4)^{-\frac{2}{3}}$$

Not derivable at x = –4

(D) $f\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$

$$f\left(\frac{\pi^-}{2}\right)=\lim_{x\to\frac{\pi^-}{2}}\left(-\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right)+\frac{\pi}{2}\right)$$

$$=\frac{\pi}{2}$$

$$f\left(\frac{\pi^+}{2}\right)=\lim_{x\to\frac{\pi^+}{2}}\left(\sin^{-1}(\sin x)\right)=\frac{\pi}{2}$$

$\Rightarrow$ f(x) is continuous at $x=\frac{\pi}{2}$

$$f(x)=\begin{cases}-\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right)+\frac{\pi}{2} & 0<x\le\frac{\pi}{2}\\ \pi-x & \frac{\pi}{2}<x<\frac{3\pi}{2}\end{cases}$$

$$f'(x)=\begin{cases}-\frac{\pi}{2} & 0<x\le\frac{\pi}{2}\\ -1 & \frac{\pi}{2}<x<\frac{3\pi}{2}\end{cases}$$

$$f'\left(\frac{\pi^-}{2}\right)=f'\left(\frac{\pi^+}{2}\right)$$

$\Rightarrow$ f(x) is differentiable at $x=\frac{\pi}{2}$

D05/5/323. Match the following column :

Column-I	**Column-II**
(a) Number of points where the function $f(x)=\begin{cases}1+\left[\cos\frac{\pi x}{2}\right] & 1<x\le 2\\ 1-\{x\} & 0\le x<1\\ \lvert\sin\pi x\rvert & -1\le x<0\end{cases}$ and $f(1)=0$ is continuous but non differentiable where [ ] denote greatest integer and { } denote fractional part function	(p) 0
(b) $f(x)=\begin{cases}x^2e^{\frac{1}{x}}, & x\ne 0\\ 0, & x=0\end{cases}$, then $f'(0^-)$ is equal to	(q) 1
(c) The number of points at which $g(x)=\dfrac{1}{1+\frac{2}{f(x)}}$ is not differentiable where $f(x)=\dfrac{1}{1+\frac{1}{x}}$, is	(r) 2
(d) Number of points where tangent does not exist for the curve $y=\text{sgn}(x^2-1)$	(s) 3

निम्न को सुमेलित कीजिए :

स्तम्भ-I **स्तम्भ-II**

(a) बिन्दुओं की संख्या जहाँ फलन $f(x)=\begin{cases}1+\left[\cos\frac{\pi x}{2}\right] & 1<x\le 2\\ 1-\{x\} & 0\le x<1\\ |\sin \pi x| & -1\le x<0\end{cases}$ (p) 0

तथा f(1) = 0 सतत् हैं लेकिन अवकलनीय नहीं जहाँ [ ] महत्तम पूर्णांक फलन तथा { } भिन्नात्मक भाग फलन को दर्शाता है

(b) $f(x)=\begin{cases}x^2e^{\frac{1}{x}}, & x\ne 0\\ 0, & x=0\end{cases}$, तब $f'(0^-)$ बराबर है (q) 1

(c) बिन्दुओं की संख्या जिन पर $g(x)=\dfrac{1}{1+\dfrac{2}{f(x)}}$ अवकलनीय नहीं है (r) 2

जहाँ $f(x)=\dfrac{1}{1+\dfrac{1}{x}}$, है

(d) बिन्दुओं की संख्या जिन पर किसी वक्र $y = \text{sgn}(x^2 - 1)$ के लिए स्पर्श रेखा विद्यमान नहीं है (s) 3

(A) $A \to q, B \to p, C \to s, D \to r$ (B) $A \to p, B \to q, C \to r, D \to s$

(C) $A \to r, B \to s, C \to q, D \to p$ (D) $A \to p, B \to s, C \to q, D \to r$

Ans. A

Sol. (A) $f(x)=\begin{bmatrix}1-1=0 & ; & 1<x\le 2\\ 0 & ; & x=1\\ 1-x & ; & 0\le x<1\\ -\sin\pi x & ; & -1\le x<0\end{bmatrix}$

at x = 0, f(x) is not continuous & not differentiable

at x = 1, f(x) is continuous & not differentiable

at x = 2 and –1, f(x) is continuous & differentiable

(B) LHD = $f'(0^-) = \lim\limits_{h\to 0}\dfrac{f(0-h)-f(0)}{-h}$

$= \lim\limits_{h\to 0}\dfrac{h^2e^{-\frac{1}{h}}-0}{h}$

$$= \lim_{h \to 0} \left(-h.e^{-\frac{1}{h}} \right) = 0$$

Ans. = Option (P)

(C) $f(x) = \dfrac{x}{x+1}$, not defined at $x = -1$

$$g(x) = \frac{f(x)}{f(x)+2}$$

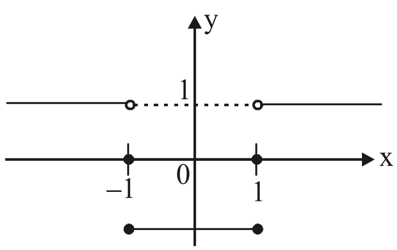
g(x) is not defined at f(x) = –2

$$\frac{x}{x+1} = -2 \Rightarrow x = \frac{-2}{3}$$

Also x = 0 is not in the domain of f(x)

So, at 3 points g(x) is not differentiable.

(D) $y = \begin{cases} 0 & ; \quad x = \pm 1 \\ -1 & ; \quad |x| < 1 \\ 1 & ; \quad |x| > 1 \end{cases}$



not continuous at x = 1, –1

$\therefore$ not differentiable at x = 1, –1

so no tangent exist at x = 1, –1 (2 points)

Ans. = option (r)

D03/5/324 Note : [x] and {x} denotes the largest integer less than or equal to x and fractional part of x respectively.

Column-I	Column-II
(a) f(x) = (x + 1) |x – 1|	(p) continuous in (–1,1)
(b) g(x) = min.(|x|, 1 – |x|)	(q) differentiable in (–1,1)
(c) h(x) = {x} + 2[x]	(r) not differentiable for at least one point in (–1,1)
	(s) discontinuous at one points in (–1,1)

नोट : [x] ≤ x तथा {x}, x के भिन्नात्मक भाग फलन को दर्शाता है।

Column-I	Column-II
(a) f(x) = (x + 1) |x – 1|	(p) (–1,1) में सतत्

(b) $g(x) = \min.(|x|, 1-|x|)$ (q) $(-1,1)$ में अवकलनीय

(c) $h(x) = \{x\} + 2[x]$ (r) $(-1,1)$ में कम से कम एक बिन्दु के लिए अवकलनीय नहीं

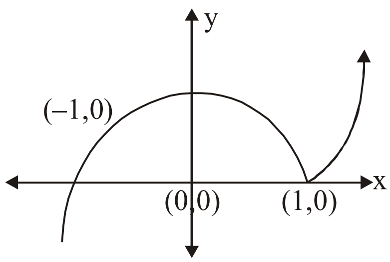
(s) $(-1,1)$ में एक बिन्दु पर असतत्

(A) $a \to s, q;\ b \to s, p;\ c \to p, r$ (B) $a \to s;\ b \to q, s;\ c \to p,q,r$

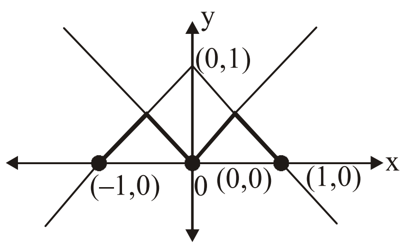
(C) $a \to r, s;\ b \to q, s;\ c \to p,r$ (D) $a \to p, q;\ b \to p, r;\ c \to r, s$

Ans. D

Sol. (A) $f(x) = (x+1)\,|x-1| = \begin{cases} x^2 - 1; & x \geq 1 \\ 1 - x^2; & x < 1 \end{cases}$



(B) $f(x) = \min.(|x|, 1-|x|)$



(c) $f(x) = \{x\} + 2[x] = x + [x]$

At $x \in I$

L.H.L. $= I + (I-1) = 2I - 1$

R.H.L. $= I + I = 2I = f(I)$

$\therefore$ Not continuous, hence not differentiable at integral points.

D03/5/325. Match the following :

Column-I **Column-II**

(a) Let $f(x) = \begin{cases} \sin x, & x \leq 0 \\ \left[\dfrac{x}{3}\right], & 0 < x < 2\pi \\ \cos x, & 2\pi \leq x < 3\pi \\ -1, & x \geq 3\pi \end{cases}$ then number of points where f(x) (p) 1

is discontinuous in $(-\infty, \infty)$, is equal to

[Note : [k] denote greatest integer less than or equal to k]

(b) If $\lim_{x\to 0}\frac{\ell n\left(1+x+x^2+....+x^n\right)}{nx}$ exists and is equal to $\frac{1}{5}$ then the value of n, is equal to — (q) 2

(c) Let g(x) = $|4x^3 - x| \cos(\pi x)$ then number of points where g(x) is non-differentiable in $(-\infty, \infty)$, is equal to — (r) 3

(d) Let f be a differentiable function such that $f'(2) = \frac{1}{4}$ then — (s) 4

$$\lim_{h\to 0}\left(\frac{f\left(2+3h^4\right)-f\left(2-5h^4\right)}{h^4}\right)$$ is equal to

(t) 5

Choose the correct option :

निम्न को सुमेलित कीजिए :

स्तम्भ-I — **स्तम्भ-II**

(a) माना $f(x)=\begin{cases}\sin x, & x\le 0\\ \left[\frac{x}{3}\right], & 0<x<2\pi\\ \cos x, & 2\pi\le x<3\pi\\ -1, & x\ge 3\pi\end{cases}$ तब $(-\infty, \infty)$ में उन बिन्दुओं की संख्या जहाँ f(x) असतत् है, बराबर होगी [नोट : $[k] \le k$] — (p) 1

(b) यदि $\lim_{x\to 0}\frac{\ell n\left(1+x+x^2+....+x^n\right)}{nx}$ विद्यमान है तथा $\frac{1}{5}$ के बराबर है तब n का मान बराबर होगा — (q) 2

(c) माना g(x) = $|4x^3 - x| \cos(\pi x)$, $(-\infty, \infty)$ में उन बिन्दुओं की संख्या जहाँ g(x) अवकलनीय नहीं है, बराबर होगी — (r) 3

(d) माना f एक अवकलनीय फलन इस प्रकार है कि $f'(2) = \frac{1}{4}$ तब — (s) 4

$$\lim_{h\to 0}\left(\frac{f\left(2+3h^4\right)-f\left(2-5h^4\right)}{h^4}\right)$$ बराबर होगा

(t) 5

सही विकल्प का चयन कीजिए :

(A) $a \to r; b \to t; c \to p; d \to q$ (B) $a \to r; b \to p; c \to s; d \to q$

(C) $a \to p; b \to q; c \to t; d \to s$ (D) $a \to r; b \to s; c \to q; d \to t$

Ans. A

Sol. (A) $f(x)=\begin{cases}\sin x, & x \ge 0 \\ \left[\frac{x}{3}\right], & 0 < x < 2\pi \\ \cos x, & 2\pi \le x < 3\pi \\ -1, & x \ge 3\pi\end{cases}$

Clearly f(x) is discontinuous at $3,6,2\pi \Rightarrow 3$ points

(B) $\lim\limits_{x\to 0} \frac{\ell n\left(1+x+x^2+....+x^n\right)}{\left(x+x^2+....+x^n\right)} \cdot \frac{\left(x+x^2+....+x^n\right)}{nx}$

$=1 \cdot \lim\limits_{x\to 0} \frac{1+x+x^2+....+x^{n-1}}{n}=\frac{1}{n}=\frac{1}{5}$

$\Rightarrow n=5$

(C) $g(x) = |x(2x + 1)(2x - 1)| \cos(\pi x)$

So, g(x) is differentiable at $x=\frac{-1}{2},\frac{1}{2}$

g(x) is continuous at x = 0, but non-differentiable at x = 0,

$\Rightarrow$ Number of points equal 1.

(D) $\lim\limits_{h\to 0}\left(\frac{f\left(2+3h^4\right)-f\left(2-5h^4\right)}{3h^4}\right)$

$=\lim\limits_{h\to 0}\left(3\left(\frac{f\left(2+3h^4\right)-f(2)}{3h^4}\right)\right)+5\left(\frac{f\left(2-5h^4\right)-f(2)}{-5h^4}\right)$

$=8f'(2)=8\left(\frac{1}{4}\right)=2$

D02/5/326. Match Column-I with Column-II and select the correct answer using the code given below the list. :

Column-I	**Column-II**
(a) If $f:[2,\infty)\to(-\infty,4]$, where $f(x)=x(4-x)$, then $f^{-1}(3)$ is equal to	(p) 0

(b) Given $g(x)=\ell n\left(a^2-a-1\right)|\cos 2x|+\left[\frac{a^3}{3}\right]\sin(3\pi x)\forall x\in R$ (q) 2

is a periodic function with period to be a rational number,
then a can be
(where [.] denotes greatest integer function)

(c) Minimum value of $h(x)=|x+1|+|x-2|+|x-3|$ is (r) 3

(d) Number of solution of equation $1+\sin x=-e^{-x}$ is (s) 4

स्तम्भ-I का स्तम्भ-II से मिलान कीजिए तथा सूचियों के नीचे दिए कूट से सही उत्तर का चयन करें :

स्तम्भ-I **स्तम्भ-II**

(a) यदि $f:[2,\infty)\to(-\infty,4]$, जहाँ $f(x)=x(4-x)$, (p) 0
तब $f^{-1}(3)$ बराबर होगा

(b) दिया गया है $g(x)=\ell n\left(a^2-a-1\right)|\cos 2x|+\left[\frac{a^3}{3}\right]\sin(3\pi x)\forall x\in R$ (q) 2

एक आवर्ती फलन है जिसका आवर्तकाल एक परिमेय संख्या है, तब a का मान
हो सकता है (जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है)

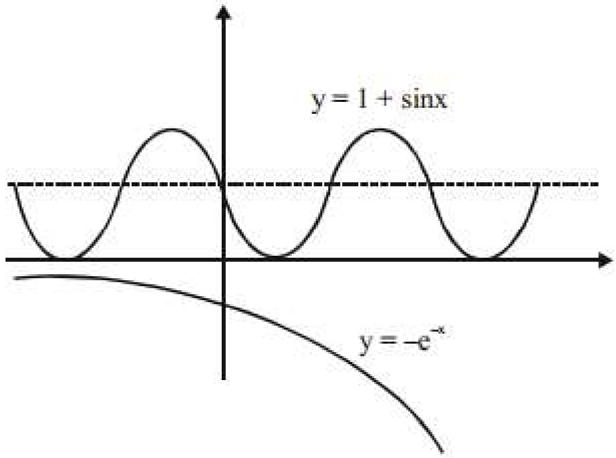
(c) $h(x)=|x+1|+|x-2|+|x-3|$ का न्यूनतम मान होगा (r) 3

(d) समीकरण $1+\sin x=-e^{-x}$ के हलों की संख्या होगी (s) 4

(A) a – r; b – q; c – s; d – p (B) a – s; b – r; c – p; d – q
(C) a – p; b – q; c – s; d – r (D) a – r; b – p; c – s; d – q

Ans. A

Sol. (a) Put $4x-x^2=3\Rightarrow x^2-4x+3=0$
$\Rightarrow x=1,3$ but $x\neq 1\Rightarrow f^{-1}(3)=3.$
(b) $\ln(a^2-a-1)$ must be zero.
$a^2-a-1=1\Rightarrow(a-2)(a+1)=0$
$\Rightarrow a=2,-1$ but $a\neq-1$ $a\neq-1$ (according to options)
$\therefore$ a = 2 and period $=\frac{2\pi}{3\pi}=\frac{2}{3}$
(c) minimum value occurs at median of –1,2,3 i.e. at x = 2
(d) By graph, no solution



D02/5/327. Match Column-I with Column-II and select the correct answer using the code given below the list :

Column-I	Column-II
(a) $f : R \to R$ and $f(x) = 2x^3 + 3x^2 + 6x$ is	(p) one one & onto
(b) $f : R \to [0, 1)$ and $f(x) = \dfrac{\|\sin x\|}{1+\|\sin x\|}$ is	(q) many one & onto
(c) $f : R \to \left(\dfrac{5}{2}, 3\right]$ and $f(x) = \dfrac{3+2\{x\}}{1+\{x\}}$ is	(r) one one & into
(d) $f : R - \left\{1, \dfrac{3}{2}\right\} \to R - \left\{\dfrac{1}{2}\right\}$ and $f(x) = \dfrac{x^2-3x+2}{2x^2-5x+3}$ is	(s) many one & into

Choose the correct option :

निम्न कॉलम का सुमेलित कीजिए :

Column-I	Column-II
(a) $f : R \to R$ तथा $f(x) = 2x^3 + 3x^2 + 6x$ है	(p) एकैकी तथा आच्छादक
(b) $f : R \to [0, 1)$ तथा $f(x) = \dfrac{\|\sin x\|}{1+\|\sin x\|}$ है	(q) बहुएकैकी तथा आच्छादक
(c) $f : R \to \left(\dfrac{5}{2}, 3\right]$ तथा $f(x) = \dfrac{3+2\{x\}}{1+\{x\}}$ है	(r) एकैकी तथा अन्तःक्षेपी
(d) $f : R - \left\{1, \dfrac{3}{2}\right\} \to R - \left\{\dfrac{1}{2}\right\}$ तथा $f(x) = \dfrac{x^2-3x+2}{2x^2-5x+3}$ है	(s) बहुएकैकी तथा अन्तःक्षेपी

सही विकल्प का चयन कीजिए :

(A) a – p, b – s, c – q, d – r
(B) a – q, b – s, c – p, d – r
(C) a – p, b – r, c – s, d – q
(D) a – s, b – q, c – p, d – r

Ans. A

Sol. (a) $f'(x) = 6x^2 + 6x + 6 > 0$

(one-one & onto)

(b) $f'(0) = f(\pi) \Rightarrow$ many one

$$f(x) = 1 - \frac{1}{1+|\sin x|} \in \left[0, \frac{1}{2}\right]$$

into function

(c) $f(0) = f(1) = 3 \Rightarrow$ many-one

$$f(x) = 2 + \frac{1}{1+\{x\}} \in \left(\frac{5}{2}, 2\right]$$

onto function

(d) $f(x) = \dfrac{(x-1)(x-2)}{(2x-3)(x-1)}$

$$= \frac{(x-2)}{(2x-3)} \; x \neq 1$$

range $= R \sim \left\{\frac{1}{2}, 1\right\}$

one-one & into

D03/5/328. Match the following column :

List-I	List-II
(a) The number of integral values of k for which the equation $\sin^{-1} x^2 + \tan^{-1} x^2 = 2k+1$ has a solution	(p) 1
(b) Let $\sum_{K=1}^{\infty} \cot^{-1}\left(\frac{k^2}{8}\right) = \frac{p}{q}\pi$ where $\frac{p}{q}$ is rational in its lowest form, then find $\lvert q-p \rvert$	(q) 2
(c) If $\tan^{-1}\left(\sin^2\theta - 2\sin\theta + 3\right) + \cot^{-1}\left(5^{\sec^2 y} + 1\right) = \frac{\pi}{2}$, then the value of $\cos^2 y - \sin\theta$	(r) 3
(d) Minimum positive integral value of x such that f(x) is defined, $f(x) = \sqrt{\sec^{-1}\left(\frac{1-\lvert x \rvert}{2}\right)}$	(s) 0

निम्नलिखित सूचियों को सुमेलित कीजिए :

सूची-I	सूची-II
(a) k के पूर्णांक मानों की संख्या जिसके लिए समीकरण का $\sin^{-1}x^2+\tan^{-1}x^2=2k+1$ का एक हल होगा	(p) 1
(b) माना $\sum_{K=1}^{\infty}\cot^{-1}\left(\frac{k^2}{8}\right)=\frac{p}{q}\pi$ जहाँ $\frac{p}{q}$ अपने निम्नतम रूप में परिमेय संख्या है तब |q – p| का मान होगा	(q) 2
(c) यदि $\tan^{-1}\left(\sin^2\theta-2\sin\theta+3\right)+\cot^{-1}\left(5^{\sec^2 y}+1\right)=\frac{\pi}{2}$, तब $\cos^2 y-\sin\theta$ का मान होगा	(r) 3
(d) x का न्यूनतम धनात्मक पूर्णांक मान होगा जिसके लिए f(x) परिभाषित है , $f(x)=\sqrt{\sec^{-1}\left(\frac{1-\|x\|}{2}\right)}$	(s) 0

(A) a – r; b – p; c – q; d – q (B) a – d; b – r; c – p; d – s

(C) a – p; b – r; c – q; d – r (D) a – p; b – s; c – q; d – q

Ans. C

Sol. (a) Let $f(x)=\sin^{-1}x^2+\tan^{-1}x^2$ is even function with domain is [–1, 1]

$$=\begin{cases}\text{decreasing in } [-1,0]\\ \text{increasing } [0,1]\end{cases}$$

$\Rightarrow f(x)|\min = f(0) = 0$

$$f(x)\big|_{\max}=f(1)=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}=f(-1)$$

$$f(x)\in\left[0,\frac{3\pi}{4}\right]$$

If k = 0, 2k + 1 = 1, possible

If k = 1, $2k+1=3>\frac{3\pi}{4}$ Not possible

i.e. only one integral value of k is possible.

(b) $T_k=\tan^{-1}\left(\frac{8}{k^2}\right)=\tan^{-1}\frac{2}{\frac{k^2}{4}}$

$$=\tan^{-1}\frac{2}{1+\left(\frac{k}{2}-1\right)\left(\frac{k}{2}+1\right)}$$

$$T_k = \tan^{-1}\left(\frac{k}{2}+1\right) - \tan^{-1}\left(\frac{k}{2}-1\right)$$

$$T_1 = \tan^{-1}\frac{3}{2} - \tan^{-1}\left(\frac{-1}{2}\right)$$

$$T_2 = \tan^{-1}\frac{4}{2} - \tan^{-1} 0$$

$$T_3 = \tan^{-1}\frac{5}{2} - \tan^{-1}\frac{1}{2}$$

$$T_4 = \tan^{-1}\frac{6}{2} - \tan^{-1}\frac{2}{2}$$

$$T_5 = \tan^{-1}\frac{7}{2} - \tan^{-1}\frac{3}{2}$$

:

:

Sum =

$$2\pi - \left(-\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{2} + \tan^{-1} 1\right) = \frac{7\pi}{4} \Rightarrow p = 7, q = 4$$

(2π is sum of last four term)

(c) we know $\tan^{-1} x + \cot^{-1} y = \frac{\pi}{2} \Rightarrow x = y$

according to question

$$\underbrace{(\sin\theta - 1)^2 + 2}_{[2,6]} = \underbrace{5\sec^2 y + 1}_{\geq 6}$$

Only possible $\Rightarrow \sin\theta - 1 = -2$

$\sin\theta = -1$

$\sec^2 y = 1 \Rightarrow \cos^2 y = 1$

$\cos^2 y - \sin\theta = 2$

(d) $\sec^{-1}\left(\frac{1-|x|}{2}\right) \geq 0$ always true in

domain $\left\{\because \sec^{-1} x \in [0, \pi] - \left\{\frac{\pi}{2}\right\}\right\}$

Now, according to domain $\frac{1-|x|}{2} \geq 1$ or

$\frac{1-|x|}{2} \le -1$

$1-|x| \ge 2$ or $3 \le |x|$

$-1 \ge |x|$ or $x \in (-\infty,-3] \cup [3,\infty)$

Not possible

least positive integer is 3

D09/5/329. Consider a circle with unit radius whose chord AB is at a perpendicular distance x from centre C. Let area of triangle ACB is given by function $y = f(x)$. On the curve $y = f(x)$ a variable point $P(x, f(x))$ is taken and perpendiculars PQ and PR are drawn on x & y axis respectively. Area of rectangle QPRO is given by $y = g(x)$. (where 'O' is origin).

Column-I	**Column-II**
(a) The greatest area of ΔACB is	(p) $\frac{2}{3\sqrt{3}}$
(b) The greatest area of rectangle QPRO is	(q) $\frac{\sqrt{13}-1}{6}$
(c) The greatest vertical distance between $y = f(x)$ & $y = g(x)$ occurs at x equal to	(r) $\frac{1}{3}$
(d) The area bounded by $y = f(x)$, $y = 0$ & $x = 1$ is	(s) $\frac{1}{2}$

Choose the correct option :

माना इकाई त्रिज्या का वृत्त जिसकी जीवा AB केन्द्र C से लम्बवत् दूरी x पर स्थित है। माना त्रिभुज ACB का क्षेत्रफल फलन $y = f(x)$ द्वारा दिया जाता है। वक्र $y = f(x)$ पर एक चर बिन्दु $P(x, f(x))$ लिया जाता है तथा x व y अक्ष पर लम्ब क्रमशः PQ तथा PR डाले जाते है। आयत QPRO का क्षेत्रफल $y = g(x)$ द्वारा दिया जाता है (जहाँ 'O' मूल बिन्दु है)

निम्न कॉलम का सुमेलित कीजिए :

स्तम्भ-I	**स्तम्भ-II**
(a) त्रिभुज ACB का अधिकतम क्षेत्रफल होगा	(p) $\frac{2}{3\sqrt{3}}$
(b) आयत QPRO का अधिकतम क्षेत्रफल होगा	(q) $\frac{\sqrt{13}-1}{6}$
(c) $y = f(x)$ तथा $y = g(x)$ के मध्य अधिकतम उर्ध्वाधर दूरी प्राप्त होगी जब x का मान होगा	(r) $\frac{1}{3}$

(d) $y = f(x)$, $y = 0$ तथा $x = 1$ द्वारा परिबद्ध क्षेत्रफल होगा (s) $\frac{1}{2}$

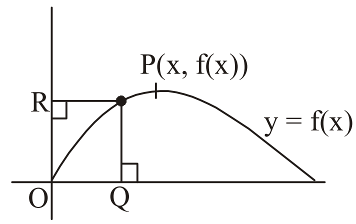
सही विकल्प का चयन कीजिए :

(A) $a \to p; b \to s; c \to q; d \to r$ (B) $a \to p; b \to q; c \to s; d \to r$

(C) $a \to p; b \to q; c \to r; d \to s$ (D) $a \to s; b \to p; c \to q; d \to r$

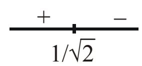
Ans. D

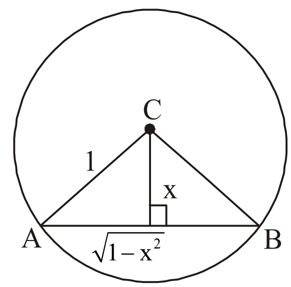
Sol. a. $f(x) = x\sqrt{1-x^2}, 0 \le x \le 1$



$$f'(x) = \frac{(1-\sqrt{2}x)(1+\sqrt{2}x)}{\sqrt{1-x^2}}$$

$f_{max}(x) = \frac{1}{2}$. Range $\left[0, \frac{1}{2}\right]$

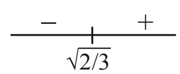




b. $g(x) = xf(x)$

$$g(x) = x^2\sqrt{1-x^2}$$

$$g'(x) = \frac{x(2-3x^2)}{\sqrt{1-x^2}}$$



Range $\left[0, \frac{2}{3\sqrt{3}}\right]$

c. $\because f(x) \ge g(x)$

$\therefore$ Vertical distance

$$h(x) = f(x) - g(x) = \sqrt{1-x^2}\,(x - x^2)$$

$$h'(x) = \frac{(1-x)(1-x-3x^2)}{\sqrt{1-x^2}}$$

$\Rightarrow$ Greatest value occurs at $x = \frac{\sqrt{13}-1}{6}$

d. Required Area $= \int_0^1 x\sqrt{1-x^2}\,dx = \frac{1}{3}$

D09/5/330. Match the following column :

Column-I | **Column-II**

(a) If α, β, γ are roots of $x^3 - 3x^2 + 2x + 4 = 0$ and (p) 2

$$y = 1 + \frac{\alpha}{x-\alpha} + \frac{\beta x}{(x-\alpha)(x-\beta)} + \frac{\gamma x^2}{(x-\alpha)(x-\beta)(x-\gamma)}$$

then value of y at $x = 2$ is

(b) If $x^3 + ax + 1 = 0$ and $x^4 + ax + 1 = 0$ have a common roots then the value of $|a|$ can be equals to (q) 3

(c) The number of local maximas of the function $x^2 + 4\cos x + 5$ is more than (r) 4

(d) If $f(x) = 2|x|^3 + 3x^2 - 12|x| + 1$, where $x \in [-1, 2]$ then greatest value of f(x) is more than. (s) 5

(t) 0

निम्न कॉलम को सुमेलित कीजिए :

स्तम्भ-I | **स्तम्भ-II**

(a) यदि α, β, γ, $x^3 - 3x^2 + 2x + 4 = 0$ के मूल हैं तथा (p) 2

$$y = 1 + \frac{\alpha}{x-\alpha} + \frac{\beta x}{(x-\alpha)(x-\beta)} + \frac{\gamma x^2}{(x-\alpha)(x-\beta)(x-\gamma)}$$

तब y का $x = 2$ पर मान होगा

(b) यदि $x^3 + ax + 1 = 0$ तथा $x^4 + ax + 1 = 0$ में एक उभयनिष्ठ मूल है तो $|a|$ का मान हो सकता है (q) 3

(c) फलन $x^2 + 4\cos x + 5$ के स्थानीय उच्चिष्ठों की संख्या अधिक होगी (r) 4

(d) यदि $f(x) = 2|x|^3 + 3x^2 - 12|x| + 1$, जहाँ $x \in [-1, 2]$ तो f(x) का अधिकतम मान अधिक होगा (s) 5

(t) 0

(A) $a \rightarrow p$; $b \rightarrow p$; $c \rightarrow t$; $d \rightarrow p, q, r, t$

(B) $a \rightarrow q$; $b \rightarrow r$; $c \rightarrow p$; $d \rightarrow s$

(C) $a \rightarrow r$; $b \rightarrow s$; $c \rightarrow q$; $d \rightarrow t$

(D) $a \rightarrow q,s,t$; $b \rightarrow r,s$; $c \rightarrow p, q, r$; $d \rightarrow s$

Ans. A

Sol. **(a)** **(p)**

$$y = \frac{x^3}{(x-\alpha)(x-\beta)(x-\gamma)}$$

$$y = \frac{8}{(2-\alpha)(2-\beta)(2-\gamma)}$$

$x^3 - 3x^2 + 2x + 4 = (x - \alpha)(x - \beta)(x - \gamma)$

put x = 2 $\quad 8 - 12 + 4 + 4 = (2 - \alpha)(2 - \beta)(2 - \gamma) = 4$

$\therefore\ y|_{\text{at } x = 2} = \frac{8}{4} = 2$

(b) (p)

$x^3 + ax + 1 = 0,\ x^4 + ax + 1 = 0$ ...(1)

$x^4 + ax^2 + x = 0$ ...(2)

(2) – (1) gives $\quad ax^2 - ax + x - 1 = 0$

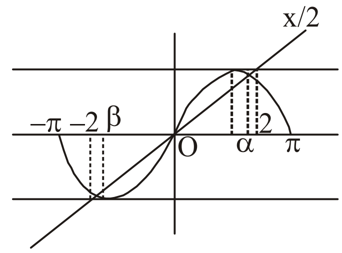
$ax(x - 1) + (x - 1) = 0$

$x = 1\ (ax + 1) = 0$

$x = 1$ or $x = -1/a$

put x = 1 in (1) weget $1 + a + 1 = 0 \Rightarrow a = -2$

$|a| = \mathbf{2}$



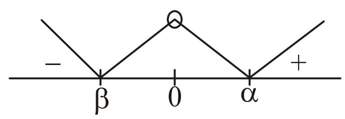
(c) (t)

$f(x) = x^2 + 4\cos x + 5$

$f'(x) = 2x - 4\sin x = 2(x - 2\sin x) = 0$

$\sin x = x/2$

$$f'(x)\begin{cases} (>0) & \text{when } x > \alpha \\ <0 & \text{when } 0 < x < \alpha \\ >0 & \text{when } \beta < x < 0 \\ <0 & \text{when } x < \beta \end{cases}$$



x = 0 only a point of maxima

so number of local maximas is **1**

(d) (p, q, r, t)

let $|x| = t \in [0, 2]$

$\therefore\ f(x) = 2t^3 + 3t^2 - 12t + 1 = g(t)$

$g'(t) = 6t^2 + 6t - 12t$

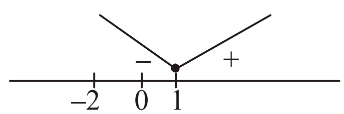
$= 6(t^2 + t - 2) = 6(t + 2)(t - 1)$

g(1) is min of f(x) i.e. $f_{min} = 2 + 3 - 12 + 1 = -6$

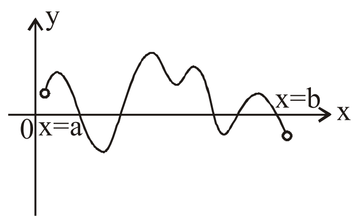
$g(0) = 1,\ g(2) = 16 + 12 - 24 + 1 = 5$

max. of f(x) is $5 > 4, 3, 2, 0$

p, q, r, t



D09/5/331. Consider the curve forf $y = f'(x)$ as shown below

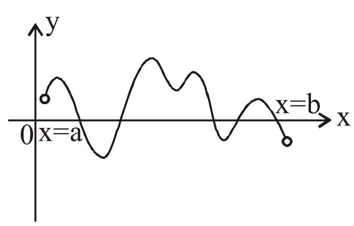


then for the curve y = f(x).

Match List-I with List-II and select the correct answer using the code given below the list.

Column-I	**Column-II**
(a) Number of points of maxima	(p) 7
(b) Number of points of minima	(q) 3
(c) Number of inflection points	(r) 5
(d) Number of stationary points	(s) 2

नीचे दिए गए वक्र f y = f ′(x) पर विचार कीजिए।



तब वक्र y = f(x) के लिए

सूची-I का सूची-II से मिलान कीजिए तथा सूचियों के नीचे दिए गए कूट के आधार पर सही उत्तर का चयन कीजिए।

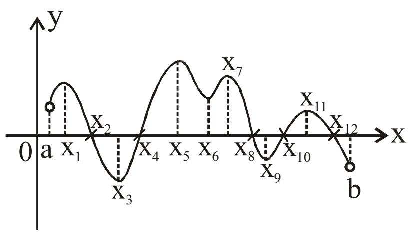
स्तम्भ-I	**स्तम्भ-II**
(a) उच्चिष्ठ बिन्दुओं की संख्या	(p) 7
(b) निम्निष्ठ बिन्दुओं की संख्या	(q) 3
(c) नति परिवर्तन बिन्दुओं की संख्या	(r) 5
(d) स्थैतिक बिन्दुओं की संख्या	(s) 2

Codes :

(A) a - r; b - s; c - p; d - q
(B) a - s; b - p; c - q; d - r
(C) a - q; b - p; c - s; d - r
(D) a - q; b - s; c - p; d - r

Ans. D

Sol.



$f'(x)$ + ve to –ve at x_2, x_8, x_{12}

$\Rightarrow$ point of maxima

$f'(x)$ –ve to +ve at x_4, x_{10}

$\Rightarrow$ point of minima

$f''(x) = 0$ at $x_1, x_3, x_5, x_6, x_7, x_9, x_{11}$

$\Rightarrow$ point of inflection

$f'(x)$ = zero at

$x_2, x_4, x_8, x_{10}, x_{12} \Rightarrow$ Stationary points

D06/5/332. **Column-I** **Column-II**

(a) If $f(x) = e^{mx}$ such that $f''(x) - 6f'(x) + 8f(x) = 0$ then the value of m is — (p) 0

(b) Number of integers in the domain of $\sqrt{\log_{\{x\}} x}$ is — (q) 2

(c) Let three sides of a triangle are three consecutive integers and largest angle is double of smallest angle, then length of largest side is equal to — (r) 4

(d) If $\sum_{r=1}^{10} \sin(rx)\cos(r^2x) = 0$, then $\frac{x}{\pi}$ can be equal to — (s) 6

(t) 8

स्तम्भ-I **स्तम्भ-II**

(a) यदि $f(x) = e^{mx}$ इस प्रकार है कि $f''(x) - 6f'(x) + 8f(x) = 0$ हो, तो m का मान होगा — (p) 0

(b) $\sqrt{\log_{\{x\}} x}$ के प्रांत में पूर्णांको की संख्या होगी — (q) 2

(c) माना त्रिभुज की तीन भुजाऐं तीन क्रमागत पूर्णाक हों तथा महतम कोण न्यूनतम कोण का दोगुना हो, तो महतम भुजा की लम्बाई होगी — (r) 4

(d) यदि $\sum_{r=1}^{10} \sin(rx)\cos(r^2x) = 0$ हो, तो $\frac{x}{\pi}$ का मान हो सकता है — (s) 6

(t) 8

(A) a $\rightarrow$ q,r; b $\rightarrow$ p; c $\rightarrow$ s; d $\rightarrow$ p,q,r,s,t
(B) a $\rightarrow$ p,s,t; b $\rightarrow$ q,r,s; c $\rightarrow$ r,t; d $\rightarrow$ p,q
(C) a $\rightarrow$ p; b $\rightarrow$ t; c $\rightarrow$ r; d $\rightarrow$ p,s,t
(D) a $\rightarrow$ r,s; b $\rightarrow$ p,q,t; c $\rightarrow$ p,q,r,t; d $\rightarrow$ q,s